



The University of Hong Kong

FITE4801 Final Year Project – Final Group Report

A Decentralized Data Wallet Unlocking Individual Data into the Next Digital Asset

Name: Luk Wing Kei (3033104368)

Wong Tsz Tung (3036065408)

Supervisor: Prof. K.H. Chow

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Abstract

Personal data has emerged as a critical digital asset, yet centralized platforms monopolize its commercial value, leaving individuals uncompensated. This final report introduces DataTrek, a Decentralized Data Wallet (DDW) designed to unlock the true value of data monetization and resolve the privacy-monetization paradox. DataTrek empowers individual data contributors to reclaim data sovereignty and earn equitable financial rewards while supplying enterprise data consumers with verifiable risk insights. This paradigm shift corrects information asymmetry, delivering profound social and economic impact by establishing a fair user-owned data economy. The project methodologies encompass a Multivocal Literature Review (MLR) to define the architectural blueprint, followed by the DataTrek Proof-of-Concept (PoC), native on-chain machine learning backend development, and Alpha and Beta testing procedures. The project successfully delivered three core components. First, the DataTrek Web Application serves as a functional proof of concept featuring an intuitive document ingestion pipeline and a gamified token rewards dashboard. Second, the on-chain machine learning (ML) algorithms utilize predictive models transpiled into Solidity Decision Trees. These algorithms execute directly on the Ethereum Virtual Machine (EVM) to generate deterministic Alternative Credit and Health Indices using gas-optimized features while completely shielding raw personal data. Third, the Evaluation Report provides a comprehensive quantitative and qualitative analysis that validates the platform's ability to democratize data value while identifying critical commercial barriers like mainnet gas fee economics, execution latency, and algorithmic opacity. Ultimately, DataTrek proves the technical feasibility of transforming private behavioral metadata into verifiable digital assets for data monetization. This establishes a trustless framework where data sovereignty and financial rewards securely coexist, providing a credible blueprint for an equitable decentralized data economy.

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List of Figures

Figure 1 Three-Token Economic Framework of DataTrek	17
Figure 2 Four-Layered System Architecture of DataTrek.....	18
Figure 3 End-to-End Operational Workflow of DataTrek.....	21
Figure 4 User Journey for Individual Data Contributors: From PDF Ingestion to Asset Liquidation	23
Figure 5 User Journey for Enterprise Data Consumers: From Data Profiles Discovery to DPI Consumptions ..	24
Figure 6 Decentralized Entry Point and MetaMask Extension Integration in DataTrek’s HomePage.....	26
Figure 7 Real-Time Connection Verification and Network Metadata.....	27
Figure 8 Logged-In Interface and MetaMask Wallet Address Display	28
Figure 9 Data Contributor’s Initial Data Upload Interface and Pre-Data Ingestion Layout.....	29
Figure 10 Active Document Ingestion and Automated Metadata Parsing.....	30
Figure 11 Data Valuation Engine Module: Real-time Analytics on Data Quality and Market Value	31
Figure 12 Privacy Governance and Data Access Permissions Toggles.....	32
Figure 13 Real-Time Enterprise Purchase Request Pop-upand Consent Verification.....	32
Figure 14 Data Consumer Portal and Marketplace Discovery Dashboard	34
Figure 15 Data Access Confirmation and Financial Breakdown Interface Upon Clicking Purchase Access to a Data Contributor’s Data Profile	35
Figure 16 Public Blockchain Audit via Sepolia Testnet Explorer	36
Figure 17 Smart Contract Escrow and Pending Approval State.....	37
Figure 18 Unlocked Comprehensive Behavioral and Health Insights Report after Successful Purchase of DPI from Individual Data Contributor's Profile.....	39
Figure 19 Dashboard: Real-Time Data Monetization and Multi-Dimensional Behavioral Analytics	41
Figure 20 Data Consumer’s API Gateway Interface for Secure Insights Retrieval for Enterprises.....	51
Figure 21 On-Chain Computation Trigger and MetaMask Execution Verification	52
Figure 22 Final Rendered On-Chain ML Insights Dashboard of a Data Contributor's DPI.....	53
Figure 23 Unified Social SSO Onboarding Interface	67
Figure 24 Web3 Wallet Onboarding and MetaMask Authorization Sequence	68
Figure 25 AI-Assisted Data Ingestion and Metadata Entry Interface	69
Figure 26 Successful Data Verification, Quality Assessment, and Market Valuation	70
Figure 27 Data Contributor-Side Data Access Request Notification	71
Figure 28 Reward Liquidation Interface – Redeem to Sepolia ETH.....	72
Figure 29 Reward Liquidation Interface – Claiming Sepolia ETH via Faucet.....	73
Figure 30 Data Buyer Portal and Data Profile Marketplace Discovery.....	75
Figure 31 Navigate to the Token DPI Access Confirmation and Financial Breakdown in Purchase Initiation...	77
Figure 32 Smart Contract Escrow and DPI Pending Approval State	78
Figure 33 Refreshed Data Consumer Interface with Unlocked Predictive DPI of an Individual’s Data Profile .	79
Figure 34 Unlocked Multi-Dimensional Behavioral Profiling and On-Chain Credit Evaluation Report	80
Figure 35 Enterprise API Gateway and Secure Authentication Portal	81
Figure 36 Pre-Execution Dashboard Displaying Encrypted DPIs.....	82
Figure 37 On-Chain Model Execution and MetaMask Smart Contract Authorization	83
Figure 38 Executed State Displaying Verified On-Chain Alternative Credit Score and Health Index	84
Figure 39 Additional Detailed Data Insights Analysis and Enterprise Risk Report generated further from DPI 85	

List of Tables

Table 1 Evaluation Measures and Methods for Beta Testing	11
Table 2 Comparative Analysis of Foundational Blockchain-based Data Infrastructures and Paradigms	13
Table 3 Tokenomics Framework in DataTrek Application	16
Table 4 DDW System Architecture Layers and Components.....	20
Table 5 On-Chain Alternative Credit Risk Stratification Tiers.....	49
Table 6 On-Chain Health & Wellbeing Index Stratification.....	50
Table 7 Credit Oracle Backtesting Metrics	54
Table 8 Health Oracle Backtesting Metrics	55
Table 9 Technical Evaluation Metrics and Outcomes in Alpha Testing.....	59
Table 10 Operational Assessment and Logistical Resilience in Alpha Testing.....	60
Table 11 Quantitative Evaluation Results and Execution Friction in Beta Testing	62

Abbreviations

API	Application Programming Interface
CIDs	Content Identifiers
DAC	Decentralized Access Control
DAO	Decentralized Autonomous Organization
dApp	Decentralized Application
DDW	Decentralized Data Wallet
DeSci	Decentralized Science
DIDs	Decentralized Identifiers
DPI	Decentralized Personal Insight
ERC	Ethereum Request for Comments (eg. ERC-20, ERC-721)
ETH	Ethereum
EVM	Ethereum Virtual Machine
GDPR	General Data Protection Regulation
HCI	Human-Computer Interaction
IPFS	InterPlanetary File System
KNN	K-Nearest Neighbors
L2	Layer-2 (Scaling Solutions)
ML	Machine Learning
MLR	Multivocal Literature Review
NFT	Non-Fungible Token
OCR	Optical Character Recognition
PoC	Proof-of-Concept
PoS	Proof-of-Stake
P2P	Peer-to-Peer
SBT	Soulbound Token
SLR	Systematic Literature Review
SSO	Single Sign-On
UCD	User-Centered Design
UI	User Interface
UX	User Experience
W3C	World Wide Web Consortium
ZKPs	Zero-Knowledge Proofs

Table of Contents

Abstract.....	i
Acknowledgement	ii
List of Figures	iii
List of Tables	iv
Abbreviations	v
1. Introduction	1
1.1 Project Background	1
1.2 Project Motivation	2
1.3 Project Objective	3
1.4 Project Scope	4
1.5 Project Deliverables	5
1.6 Report Outline	6
2. Project Methodologies	7
2.1 Multivocal Literature Review	7
2.2 Prototyping and PoC: Frontend Design and Feature Implementation	8
2.3 Backend Implementation: On-Chain Machine Learning	9
2.4 Two-Phase Evaluation: Internal Alpha Testing and User-Centered Beta Testing	10
3. Results and Discussions	12
3.1 Findings from MLR	12
3.2 System Architecture and Tokenomics Framework of DataTrek	16
3.3 System Flow and User Experience	21
3.4 Frontend Interfaces and Feature Integration	26
3.5 Native On-Chain Machine Learning Backend Implementation	42
3.6 System Evaluation and Testing Results	57
3.7 End-to-End Application Walkthrough (User Manual)	66
4. Conclusion, Limitations, and Future Work	86
5. Project Schedule	88
References	90

1. Introduction

1.1 Project Background

Personal data has emerged as a critical asset in the modern digital economy [1]. Organizations, such as the World Economic Forum, have classified digital footprints as a distinct asset class, suggesting that individuals should be able to capitalize on their own information [2]. However, the current data ecosystem is highly centralized. Technology corporations aggregate user data through proprietary infrastructures and monetize it via third-party sales and targeted advertising, usually with limited transparency and without direct financial compensation to the data creators [3], [4].

This centralized model results in significant information asymmetry and economic inequity [5]. To address this, blockchain-based solutions like DataMesh+ [6] and Ocean Protocol [7] have introduced peer-to-peer (P2P) data exchanges. While these platforms return data ownership to users, they generally focus on trading raw datasets off-chain. This approach still exposes individuals to privacy risks during data transfers and forces data consumers to process noisy and unstructured information.

1.2 Project Motivation

Despite the theoretical advantages of decentralized ownership, current data monetization solutions face significant technical limitations. This project, titled DataTrek, is motivated by the need to resolve challenges across three specific dimensions:

- 1. Flaw in Current Data Monetization:** Existing marketplaces prioritize trading raw datasets over derivable, actionable insights. They lack frameworks to tokenize insights as standardized digital assets, resulting in inefficient interoperability and limited data utility.
- 2. Problem for Data Contributors (Individuals):** Data fragmentation makes it difficult for users to control their information. Consequently, individuals are trapped in a privacy-monetization paradox: they cannot profit from their data without exposing raw, sensitive personal details to third parties.
- 3. Problem for Data Consumers (Enterprises):** Enterprises struggle with weak data provenance, leading to unreliable insights generated from polluted datasets. Furthermore, early decentralized analytics rely on off-chain backends. This introduces a "black box" execution vulnerability where consumers must blindly trust a centralized, opaque server to execute the algorithm fairly, undermining the core premise of decentralization.

To address these challenges and resolve this execution vulnerability, the project architecture was pivoted to execute machine learning (ML) inference natively on the blockchain. Because the Ethereum Virtual Machine (EVM) has strict computational and gas limitations, running standard ML models on-chain is challenging. By utilizing lightweight algorithms — specifically Decision Trees adapted for integer arithmetic - DataTrek allows raw user data to be transformed into Decentralized Personal Insights (DPIs) directly via smart contracts. This resolves the privacy-monetization paradox: it allows enterprises to access verified, standardized metrics with zero exposure to the underlying raw personal data, while data contributors are economically incentivized through automated, trustless token rewards, establishing an equitable "Contribute-to-Earn" economy.

1.3 Project Objective

The primary goal of this project is to conceptualize, design, and implement DataTrek to bridge the gap between secure data ownership and machine learning analytics through a tokenized reward system. The specific objectives are as follows to:

- **Safeguard Data Sovereignty:** Empower individuals with full authority to directly grant or deny access to their personal data insights.
- **Ensure Transparent Analysis:** Guarantee that all data processing is verifiable and trustworthy by natively executing analytical algorithms in a transparent framework.
- **Create Fair Monetization:** Develop an equitable model that directly rewards contributors based on the value, quality, and demand of the data they provide.
- **Establish Data Provenance:** Create a permanent, publicly verifiable record of data ownership while ensuring the underlying raw data remains completely private [27].

1.4 Project Scope

DataTrek is engineered as a versatile infrastructure. To ensure technical feasibility and a focused evaluation during this development cycle, the project scope has been defined across the following dimensions:

- **Supported Data Modalities:** As DataTrek is currently developed for Proof-of-Concept (PoC) purposes, the system's data ingestion is intentionally designed to accompany to unstructured PDF documents only. It utilizes Optical Character Recognition (OCR) to automatically extract behavioral metadata from travel documents and official records.

While other data formats, such as images, .csv files, and raw spatial data arrays, could be seamlessly integrated in future deployments. By constraining the initial scope to PDFs, this allows the project to place a stronger emphasis on testing and validating the complete, end-to-end user journey.

- **Data Domains:** Travel behavioral data have been selected as the primary data substrate due to their high analytical value for evaluating lifestyle stability and financial applications.
- **Application Scenarios (On-Chain ML algorithms):** The extracted metadata is processed by on-chain ML algorithms to synthesize behavioral metadata into actionable insights. Specifically, the PoC demonstrates the generation of an (1) **Alternative Credit Score** and a (2) **Health Index Score**.

This involves leveraging non-traditional data sources to assess an individual's creditworthiness, thereby promoting more inclusive lending decisions and reducing reliance on centralized financial histories [11]-[14]. Similarly, the Health Index Score utilizes spatial and travel behavioral patterns to evaluate an individual's lifestyle regularity and mobility, providing a privacy-preserving risk assessment metric for wellness and insurance applications.

1.5 Project Deliverables

This project delivers the DataTrek platform, a functional web-based PoC, alongside a comprehensive academic evaluation. To clearly distinguish the user-facing interfaces from the underlying blockchain logic, the deliverables are categorized into three components:

- **DataTrek Web Application (Frontend-implementation):**

A dual-interface decentralized application (dApp). It includes a *Data Contributor Portal* for individuals to upload documents via OCR, manage permissions, and track protocol-based rewards , and a *Data Buyer Portal* for enterprises to browse data contributors' data profiles and execute access requests.

- **On-Chain ML (Backend-implementation with Smart Contracts):**

The decentralized execution layer deployed on the Ethereum Sepolia Testnet. With smart contracts, this component governs the core logic, including token escrow mechanisms, metadata linkage, and the native on-chain execution of Decision Tree inference (adapted for integer arithmetic [29]) to generate the Alternative Credit Score and Health Index without relying on a centralized backend.

- **DataTrek's System Evaluation Report:**

An evaluative report detailing the results and findings from the alpha and beta testing phases. The report focuses specifically on the system's technical practicality and performance metrics as DataTrek transitioned from utilizing synthetic data to processing real-world datasets for its core analytical applications.

1.6 Report Outline

The remainder of the report is organized as follows: Section 2 outlines the project methodology and the multi-layered system architecture. Section 3 details the technical implementation and the end-to-end system workflow. Section 4 presents the evaluation and results from the alpha and beta testing phases. Section 5 provides an empirical analysis of the project's practicality, discussing technical limitations, business value, and future deployment capabilities. Finally, Section 6 concludes the report.

2. Project Methodologies

Section 2.1 outlines the development of the technical protocol specifications and the system architecture blueprint using Multivocal Literature Review (MLR). Section 2.2 covers the prototyping of DataTrek’s user interfaces as a dApp. Section 2.3 details the backend on-chain machine learning development. Section 2.4 details the subsequent Alpha and Beta testing procedures.

2.1 Multivocal Literature Review

A MLR, a systematic review approach that integrates both academic research and grey literature, including white papers, technical reports, standards documentation, and practitioner blogs [15], was adopted to develop the technical protocol specifications and system architecture blueprint for DataTrek. Unlike a traditional Systematic Literature Review (SLR), which relies exclusively on peer-reviewed academic literature, the MLR is particularly valuable in fast-evolving research domains like decentralized data economies, where academic research often lags behind industrial innovation [15].

The core aim of this MLR was to facilitate a deep inquiry into the practical application of blockchain technology, decentralized architectures, and current industry practices. To direct this scope of inquiry, several thematic research objectives were utilized to strengthen the development of the platform's 4-layered system architecture and its supporting computational components. These areas of investigation were intentionally dynamic throughout the research process, particularly to accommodate the project's pivot toward researching on-chain machine learning constraints and verifiable execution.

By synthesizing insights from academic databases and industry repositories, the MLR ensured that the system architecture and the protocol design of DataTrek were both theoretically robust and practically implementable. Sources will be drawn from academic databases and industry repositories, including insights produced by industrial practitioners, companies’ whitepapers, technical reports, and protocols on tokenization projects. Literature published between 2020 and 2025 is prioritized to capture the latest developments in the decentralized data economy. This comprehensive synthesis ultimately provided a verified foundation for aligning the project's technical specifications with contemporary industry standards and emerging blockchain paradigms.

2.2 Prototyping and PoC: Frontend Design and Feature Implementation

The development of DataTrek followed an evolutionary prototyping methodology, focusing on the functional validation of the system's decentralized core while iteratively refining user interactions. A functional web-based version of the prototype was developed to demonstrate end-to-end user journeys for both data contributors (individuals) and data consumers (enterprises).

To effectively separate the user experiences (UXs), the frontend was architected into two core interfaces:

- **The Data Contributor Portal:** Designed to empower individuals with data sovereignty, this interface facilitates secure off-chain data ingestion, cryptographic consent management, and the settlement of token rewards.
- **The Data Consumer Portal:** Functioning as a decentralized B2B marketplace, this interface provides an enterprise dashboard for institutional buyers to browse anonymized user profiles, submit access requests, and securely unlock generated insights via smart contract interactions.

The User Interface (UI) across both portals was meticulously designed to abstract and simplify the inherent complexities of blockchain interactions, while ensuring complete transparency for all stakeholders. By focusing on these key interactions, the PoC served as a functional testing ground to evaluate the seamless integration of React-based frontend components with the underlying blockchain logic.

Sepolia testnet on Ethereum was selected as the backend decentralized base layer that the frontend interacted with. The Ethereum Sepolia testnet provided a simulated blockchain environment that mirrored Ethereum's mainnet functionality, using valueless test Ethereum. It was chosen because it provided a cost-effective and robust environment to test the frontend's ability to reliably trigger smart contracts, handle token escrow, and execute on-chain ML inferences during the initial development stages.

2.3 Backend Implementation: On-Chain Machine Learning

The backend methodology on on-chain ML focused on the development and migration of the machine learning inference engine directly into the Ethereum smart contract layer. To achieve this within the strict computational constraints of the blockchain, the technical approach was structured into four key stages:

- **Feature Engineering:** The raw travel behavioural metadata extracted via OCR required standardization before it could be ingested by a smart contract. The methodology involved engineering these unstructured data points into discrete, quantifiable variables (features) that could be processed efficiently on-chain.
- **Algorithm Selection:** Computationally heavy machine learning models are inherently incompatible with the EVM due to strict block gas limits. To address this, various models were evaluated against Ethereum's computational constraints. Ultimately, lightweight and deterministic Decision Trees were selected as the optimal architecture for native on-chain execution. *(A detailed explanations of this algorithmic pivot from advanced ML algorithms to Decision Trees is provided in the Section 3 to highlight the technical constraints and our rationale for this selection)*
- **Mathematical Translation into Solidity:** Because the Solidity programming language natively lacks support for floating-point mathematics, the methodology required a structural adaptation of the ML algorithms. This involved designing an integer arithmetic scaling mechanism to accurately represent and compute fractional probabilities entirely within the smart contract environment.
- **Model Backtesting:** Prior to immutable deployment, the selected ML algorithms were first trained and backtested off-chain using historical datasets. This iterative testing phase was necessary to optimize the conditional split logic and ensure high accuracy for the insight generation before committing the rules to the blockchain.

2.4 Two-Phase Evaluation: Internal Alpha Testing and User-Centered Beta Testing

2.4.1 Phase I - Internal Technical and Operational Evaluation (Alpha Testing)

The PoC underwent internal Alpha Testing to confirm functional integrity prior to user-facing trials. Utilizing a controlled environment with simulated data contributors and consumers, this phase focused on verifying the system's core decentralized logic, specifically testing the new pure on-chain ML architecture. Iterative refinements were implemented based on internal performance data to pre-empt functional errors.

- **Technical Verification:** Assessed the accuracy of the data-to-asset transformation (processing raw PDF data to minted tokens), the logic of the smart contract execution (specifically the native on-chain ML inferences and compensation mechanisms), and the efficacy of cryptographic privacy mechanisms.
- **Operational Assessment:** Evaluated the layered architecture's integrity to ensure cohesive interaction between the frontend interfaces, off-chain storage (IPFS), and the blockchain execution layer, monitoring system reliability under simulated conditions.

2.4.2 Phase II - User-Centered Design (UCD) Evaluation (Beta Testing)

Following the technical refinements made, Beta Testing was conducted to assess the prototype's utility. During this phase, the project pivoted its focus toward transitioning the prototype from a theoretical PoC to a real-world practical deployment capable of processing real-world data applications.

Instead of conducting a broad quantitative survey with a 20 participants user base as stated in the interim report, the Beta testing pool is deliberately restricted to 2 highly focused participants acting as data contributors and data consumers. This methodological pivot was designed to:

1. **Maximize Qualitative Depth:** Deeply analyze how users interact with complex Web3 and on-chain mechanics, such as MetaMask signing and cryptographic consent, when utilizing their own real-world travel data rather than synthetic and simulated datasets.

- 2. Facilitate Rapid Iteration:** Allow us to make immediate and highly targeted amendments to the protocol and UI from detailed and granular feedback.

Participants completed a series of representative tasks to evaluate the system's functional performance and overall UX. Feedback was gathered and analyzed through an adapted set of metrics (see Table 1).

Evaluation	Measurement & Method
Quantitative Measures	• Task completion rate: Tracking whether the two participants successfully executed the end-to-end data pipeline. • Error rate: Identifying specific functional friction points or mistakes made during Web3 interactions.
Qualitative Insights (Primary Focus)	• Semi-structured interviews: Guided conversations to evaluate user trust, comprehension of data sovereignty, and overall experience with real-world data application. • Think-aloud protocol: Users verbalizing their thought processes while uploading documents and interacting with the smart contracts.

Table 1 Evaluation Measures and Methods for Beta Testing

***Note:** Quantitative metrics are used to evaluate task functional efficiency, accuracy, and system-wide usability., while qualitative insights aim to understand participants' user experience, perceptions regarding trust, data sovereignty, and satisfaction with the reward system in the DDW. Qualitative insights are prioritized in this phase to explore the nuances of user trust in decentralized fintech architectures, which are often more critical for adoption than raw speed.*

This UCD approach ensured that the system was designed in alignment with user expectations, particularly regarding trust and data sovereignty in real-world conditions.

3. Results and Discussions

Section 3 presents the final technical implementation and performance analysis of the DataTrek Proof-of-Concept. Section 3.1 details the Findings from the MLR, while Section 3.2 describes the System Architecture and Tokenomics Framework of DataTrek. Section 3.3 outlines the System Flow and User Experience, followed by Section 3.4 which showcases the Frontend Interfaces and Feature Integration. Section 3.5 documents the Native On-Chain Machine Learning Backend Implementation, and Section 3.6 presents the System Evaluation and Testing Results. Section 3.7 provides an End-to-End Application Walkthrough serving as functional User Manuals for both individual (data contributors) and enterprises (data consumers).

3.1 Findings from MLR

The purpose of this MLR was to synthesize insights from both academic research and grey literature to establish the foundational requirements for **DataTrek**. This review systematically reviewed and evaluated existing blockchain-based data infrastructures while gaining insights from industry trends to inform the project’s system architecture, protocol design, operational workflow and final on-chain execution logic.

3.1.1 Comparative Analysis of Blockchain Data Infrastructures

Table 2 compares the four primary paradigms and industry applications identified during the MLR. These cases represent a broad architectural spectrum, ranging from identity management to autonomous insight generation, that informed the development of DataTrek.

Case Study	Type	Spectrum Representation	What is it doing?	Core Problem Solved	Novelty / Key Contribution	Key Standards / Technical Mechanisms
Civic (Civic Pass) [19], [20]	Industry	Identity & Authentication	Issuing "passes" (represented by SBTs) to verify user identity directly on-chain.	Eliminates repetitive Know Your Customer and prevents Sybil attacks in P2P markets.	Links real-world identity to wallet addresses using SBTs [19].	World Wide Web Consortium (W3C) Decentralized Identifiers (DIDs) [20], SBTs [19]
DataMesh+ [6]	Research	Data Infrastructure & Exchange	Managing P2P data exchanges through decentralized access control (DAC).	Resolves bottlenecks in centralized IoT data marketplaces.	A hybrid P2P exchange model for high-performance self-sovereign data [6].	DAC, P2P Protocols [6]
Ocean Protocol [7], [17]	Industry	Data Monetization & Assetization	Converting data access rights into liquid financial assets (Data Tokens).	Breaks down data silos by creating a standardized monetization layer.	Tokenizes access rights to create a liquid, privacy-preserving data economy [7].	ERC-721 (Data NFTs), ERC-20 (Datatokens) [6]
Insight Genesis [18]	Industry	Artificial Intelligence (AI) & ML Application	Transforming raw datasets into verifiable, actionable insights via AI.	Solves the "data noise" problem where raw data is too complex for consumers.	Focuses on the output (insights) rather than the input, streamlining ingestion [18].	Verifiable Computation, AI Oracle integration [18]

Table 2 Comparative Analysis of Foundational Blockchain-based Data Infrastructures and Paradigms

Note: This table provides a multidimensional comparison of the primary research and industry frameworks evaluated during the MLR. Essential components are identified for building the DDW's decentralized data-to-insight pipeline for the DDW prototype.

3.1.2 Cross-Synthesis of MLR Findings across Academic and Industry Literatures

The findings from MLR revealed a fundamental architectural evolution within the decentralized data landscape: a shift from rudimentary P2P storage models toward intelligent, verifiable data ecosystems that prioritize utility alongside sovereignty. A primary finding, synthesized from the peer-reviewed work of Merlec and In [6] regarding DataMesh+ and the industry-standard protocols advocated by McConaghy [17] of Ocean Protocol, was the operational necessity of decoupling identity from raw behavioral data. This was achieved through the implementation of DIDs and SBTs, as pioneered in the "Self-Sovereign Identity" (SSI) manifesto by Christopher Allen [19] and formalized by the W3C standards [20], [28]. As highlighted by the latest technical surveys on decentralized identifiers and verifiable credentials [25], these frameworks facilitate a "trustless" verification layer where the user retains absolute ownership of their digital footprint while providing granular, permissioned access to third parties for specific computational tasks.

Furthermore, the synthesis identified a critical "utility gap" in current frameworks: while robust infrastructure exists for secure data hosting, there remained a significant deficit in systems capable of autonomously transforming high-entropy, raw personal data into structured predictive metrics or verifiable ML insights without compromising underlying privacy. The documentation for Insight Genesis [18] addressed this by shifting the focus from simple data assetization to verifiable insight generation. This conceptual shift directly informed DataTrek's internal logic, enabling a seamless pipeline that converted user-held data into high-value ML insights through native on-chain execution.

3.1.3 Contribution to DataTrek Logic and Concept Formation

Collectively, the findings from MLR mandated DataTrek's adoption of an evolving decentralized authentication model. This research directly informs the DDW's "Contribute-to-Earn" economic logic, utilizing the Web3 Data Value Chain [17] to transform the user from a passive data subject into an active participant. This mechanism ensures a sustainable equilibrium between data supply and computational demand, mirroring the "Compute-to-Data" staking models observed in Ocean Protocol [6].

To realize this, the theoretical findings were translated into a concrete two-layer technical architecture and a corresponding three-token economic framework. These finalized technical and economic components are detailed extensively in Section 3.2.

3.2 System Architecture and Tokenomics Framework of DataTrek

This sub-section outlines the finalized structural blueprint of the DataTrek protocol, detailing the economic incentive mechanisms and the multi-layered technical infrastructure engineered to support secure, decentralized data processing.

3.2.1 Protocol Logic and Tokenomics Framework

To realize the "Contribute-to-Earn" logic established during the research phase, DataTrek's finalized architecture partitioned the conceptual workflow into two distinct stages:

- **A Local Privacy Layer:** Raw personal behavioral data remains strictly off-chain. It is stored via the encrypted IPFS and pre-processed to ensure sensitive personal information never enters the public ledger or a centralized database.
- **An On-Chain Execution and Verification Layer:** Where ML inferences are natively executed within the smart contract environment. Only the derived insights, strictly excluding raw data, are validated, finalized, and minted as Data NFTs (Access Licenses) for direct consumption.

The execution of this two-layered model was facilitated by a three-token system, which codified the relationship between data ownership, insight utility, and the tokenomics framework within the DataTrek protocol (see Table 4).

Token Type	Category	Purpose	Transferability
Data Ownership Token	Soulbound Token (SBT)	Acts as a "Digital Deed" proving the user is the verified owner of the raw data [26].	Non-Transferable
Insight Token (DPI Token)	Data NFT/License	A cryptographic summary of a specific ML insight that acts as the "Access Ticket" for a consumer.	Restricted (Licensed to Payer)
Reward Token	Utility Token	The fungible currency used by enterprises to pay for access and by users to receive incentives.	Transferable & Redeemable to ETH

Table 3 Tokenomics Framework in DataTrek Application

To visualize how these digital assets operationalize the protocol's economy, Figure 1 illustrates the end-to-end interaction and directional flow of the three token types between data contributors and enterprise consumers.

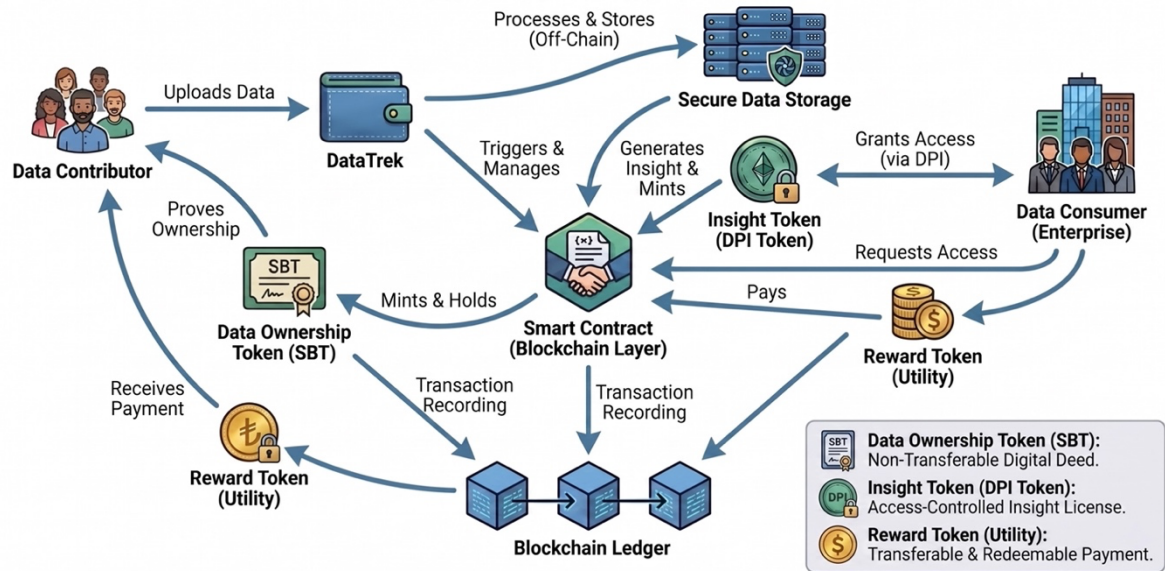


Figure 1 Three-Token Economic Framework of DataTrek

Note: This figure visualizes the decentralized transaction flow, showing how SBTs, DPI tokens, and Reward tokens mediate secure exchange between contributors and enterprises.

3.2.2 Four-Layered System Architecture Blueprint

The DataTrek system architecture is organized into a four-tier modular protocol stack comprising the Base, Core, Service, and Application Layers, complemented by a set of supporting components (see Figure 2 & Table 4).

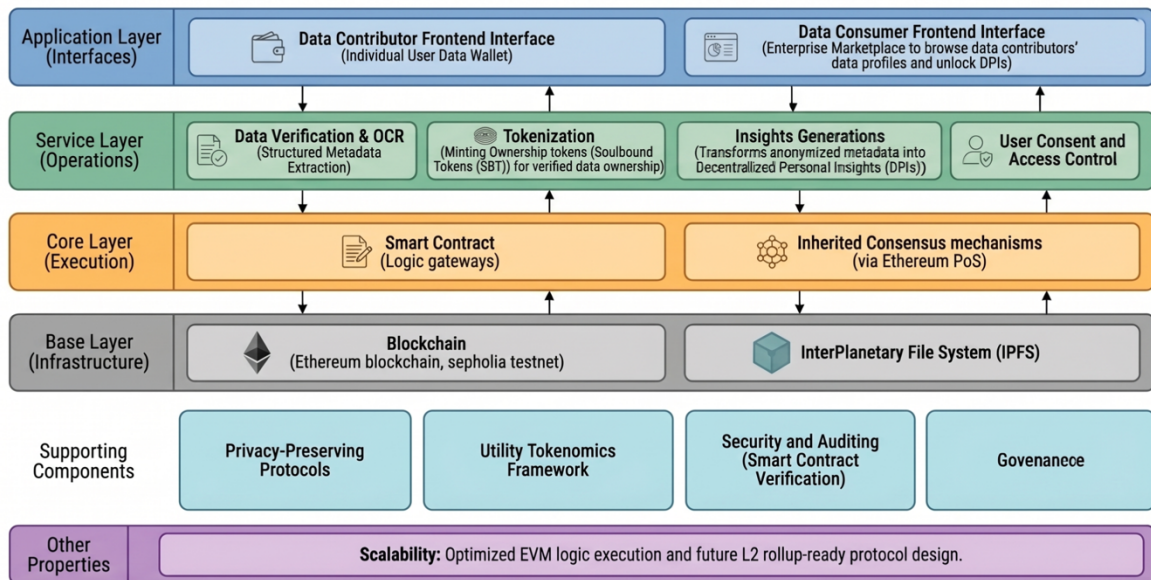


Figure 2 Four-Layered System Architecture of DataTrek

***Note:** This schematic illustrates the modular, four-tiered architecture of DataTrek, demonstrating the strict decoupling of user-facing client interfaces from the underlying execution, operational, and infrastructure layers.*

The **Base Layer** provides the foundational infrastructure of DataTrek, leveraging the Ethereum Sepolia testnet and IPFS. Ethereum serves as an immutable, tamper-proof ledger for transaction and token records, while IPFS offers low-cost, decentralized storage for encrypted raw data. This hybrid storage approach successfully offloads high-entropy files from the blockchain to enhance scalability and cost-efficiency, while maintaining a verifiable link to the original data via on-chain Content Identifiers (CIDs).

The **Core Layer** defines the operational logic and trust framework of the protocol, consisting of smart contracts and Ethereum’s Proof-of-Stake (PoS) consensus mechanism. These smart contracts act as the central computational environment within the EVM. They automatically enforce the rules of the DataTrek protocol, verifying data submissions, executing on-chain data evaluations, issuing ownership tokens, and managing granular access permissions—all without relying on a centralized off-chain server.

The ***Service Layer*** acts as the functional engine that transforms off-chain data into tokenized digital assets. Within this layer, a decentralized OCR and verification process extracts structured metadata from the uploaded records. The tokenization engine then mints non-transferable Data Ownership Tokens (SBTs) to bind data permanently to the contributor's wallet. This layer also manages the user consent mechanisms, allowing data contributors to review, approve, or reject consumer requests using precise, time-bound permissions enforced directly by the smart contracts.

The ***Application Layer*** provides the user-facing interfaces for interaction between the two primary stakeholders. The **Data Contributor Interface** serves as a decentralized portal where individuals securely upload personal data, manage granular access permissions, and track protocol-based rewards. In parallel, the **Data Consumer Interface** is an enterprise-facing marketplace that allows businesses to browse anonymized data profiles and initiate smart-contract-governed payments to unlock specific DPIs.

Beyond the four primary layers, this architectural stack is augmented by several vital ***supporting components*** to ensure security and economic viability. Privacy-Preserving Protocols maintain data anonymity by strictly keeping raw files off-chain and passing only essential, non-identifiable metadata to the EVM. This is complemented by the Utility Tokenomics Framework, which automates the system's "Contribute-to-Earn" economy via smart contracts, ensuring that reward tokens are automatically distributed to contributors the moment a data consumer unlocks an insight. The system's integrity is maintained by two final pillars. Protocol Governance is established through the hard-coded logic of the smart contracts, which act as the final authority on data access rights and reward distribution. To ensure this logic is tamper-proof, DataTrek prioritizes Security and Auditing by utilizing verified smart contract code, which provides a transparent and trustless environment for all stakeholders.

Regarding ***scalability***, DataTrek achieves a highly efficient execution model by deliberately optimizing its computational ML workloads to run within the EVM. This approach successfully avoids gas limit bottlenecks, ensuring that high-frequency token minting and on-chain logic execution remain economically viable. By utilizing this modular design, the protocol establishes a structurally resilient foundation that is fully prepared for future deployment on Ethereum Layer-2 (L2) rollup solutions, allowing the system to scale alongside growing network demands.

Layer	Components
Base Layer (Infrastructure)	• Blockchain (Ethereum Sepolia testnet) • Decentralized Storage (InterPlanetary File System (IPFS))
Core Layer (Execution)	• Smart Contracts (Logic gateways and native on-chain ML evaluation) • Consensus Mechanism (Ethereum PoS)
Service Layer (Operations)	• Tokenization Engine (Issues non-transferable SBTs for ownership and Data NFTs for insight monetization) • Automated Data Validation (Integrity checks for uploaded behavioral records) • Insight Generation Engine (Transforms processed data features into verifiable, anonymized personal insights) • User Consent & Access Control (Management of permissioned requests for viewing DPIs)
Application Layer (Interfaces)	• Data Contributor Interface (Secure data upload, consent management, and reward tracking) • Data Consumer Interface (Anonymized profile browsing and insight acquisition)
Supporting Components	• Privacy-Persevering Protocols • Utility Tokenomics Framework • Security and Auditing (Smart Contract Verification) • Protocol Governance
Other properties	• Scalability

Table 4 DDW System Architecture Layers and Components

***Note:** This table summarizes the system architectural layers of DataTrek and the key components in each layer to ensure functional clarity and operational coherence.*

3.3 System Flow and User Experience

This section details the operational mechanics of the DataTrek protocol. Section 3.3.1 outlines the systemic and end-to-end technical workflow of DataTrek, while Section 3.3.2 maps the corresponding interactive journeys from the perspective of both data contributors and data users as primary stakeholders.

3.3.1 End-to-End Operational Workflow

DataTrek's end-to-end operational workflow is illustrated in Figure 3. This workflow maps the sequential technical interactions between the off-chain extraction nodes, the on-chain execution environment, and the bilateral settlement mechanisms.

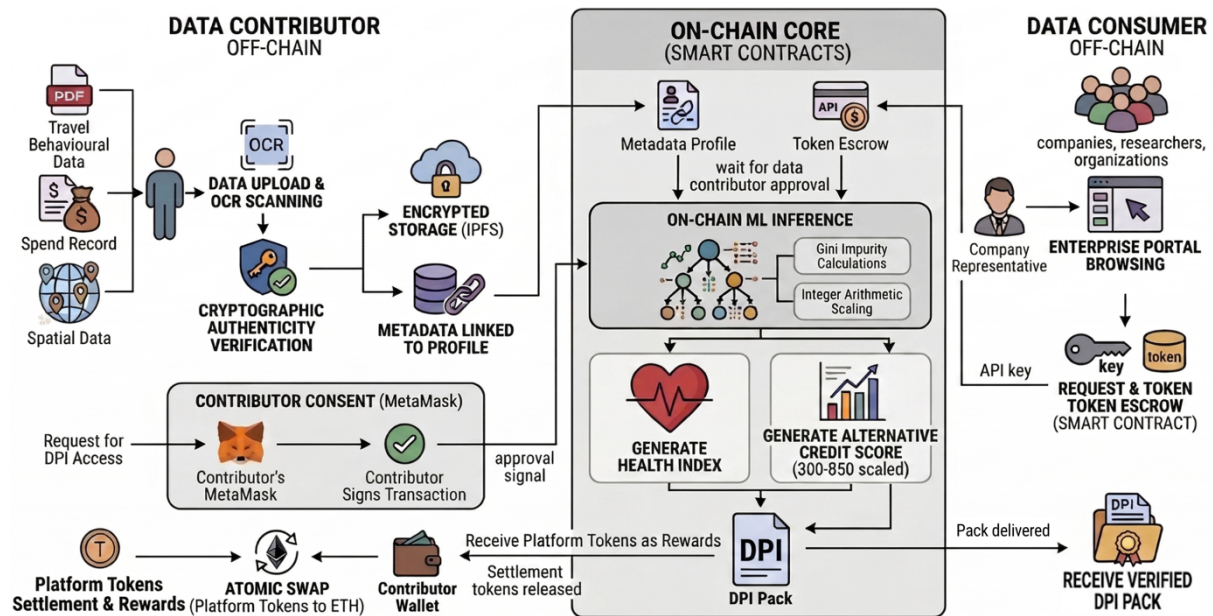


Figure 3 End-to-End Operational Workflow of DataTrek

Note: This high-level flowchart illustrates the secure data pipeline, detailing the transition from off-chain data processing to native on-chain machine learning inference and automated token settlement.

Technically, the systemic workflow operates through five chronological phases:

1. **Off-Chain Data Ingestion and Verification:** The flow begins when raw travel behavioral data (e.g., PDF spend records) is uploaded. An off-chain OCR scanning mechanism extracts relevant spatial and behavioral metrics. Cryptographic authenticity verification is performed before the encrypted raw data is sent to IPFS for storage. Only the non-identifiable metadata is linked to the user's on-chain profile.

2. **Consumer Escrow and Request:** An enterprise consumer browses available metadata profiles via one's portal. Upon identifying a relevant profile, the enterprise requests access and locks Utility Tokens into a smart contract escrow, signaling a guaranteed payment.
3. **Cryptographic Consent Routing:** The protocol routes the access request to the Data Contributor. The system pauses until the contributor explicitly signs the approval transaction using their Web3 wallet (MetaMask).
4. **On-Chain ML Inference (Core Execution):** Once the approval signal is received, the smart contract natively executes the machine learning inference. Utilizing Gini Impurity calculations and Integer Arithmetic Scaling to remain within EVM limits, the protocol processes the metadata to generate specific insights, such as a Health Index and an Alternative Credit Score.
5. **Automated Settlement and Delivery:** The smart contract automatically finalizes the transaction. The generated DPI is securely delivered to the enterprise. Simultaneously, the escrowed tokens are released directly to the contributor's wallet, concluding the loop with an optional atomic swap to ETH.

3.3.2 Dual-sided User Journeys

3.3.2.1 The Individual Data Contributor Journey

The contributor's journey is built on the principle of regaining personal data sovereignty, ensuring that every interface interaction maximizes transparency and user control.

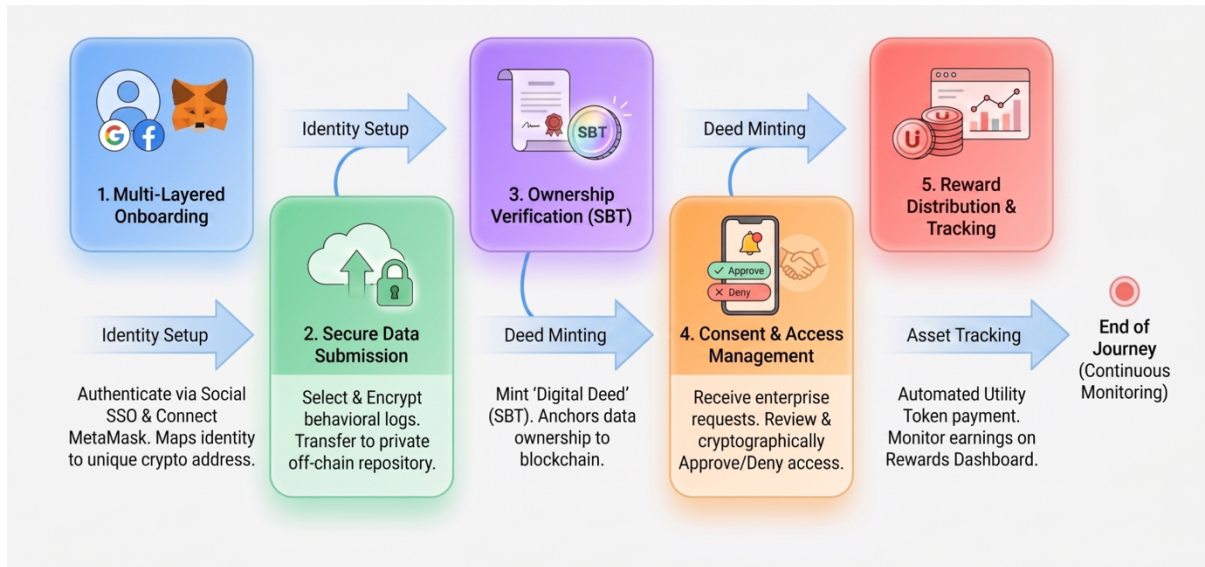


Figure 4 User Journey for Individual Data Contributors: From PDF Ingestion to Asset Liquidation

In **Stage 1**, the **Multi-Layered Onboarding stage**, the user enters the application by authenticating through a familiar Social SSO (Single Sign-On), e.g., Google or Facebook, for a low-friction entry. They are then prompted to connect their MetaMask wallet, seamlessly mapping their Web2 identity to a secure Web3 cryptographic address.

In **Stage 2**, the **Secure Data Submission stage**, the user selects specific behavioral logs to upload. The interface provides immediate visual feedback confirming that the file is encrypted locally and that raw personal data is securely transferred to a private off-chain repository, never touching the public ledger.

In **Stage 3**, the **Ownership Verification (SBT Minting) stage**, following a successful upload, the user's dashboard instantly updates to display a newly minted SBT. This acts as a "Digital Deed," providing visual confirmation that their data asset is officially anchored to the blockchain.

In **Stage 4**, the **Consent and Access Management stage**, the user receives real-time notifications when an enterprise requests their insights. The user reviews the requester's identity and the offered token reward, choosing to 'Approve' or 'Deny' via a MetaMask signature. No data processing occurs without this active consent.

In **Stage 5**, the **Reward Distribution and Asset Tracking stage**, upon a successful exchange, the user is routed to a Rewards Dashboard. Here, they receive automated Utility Token payments and can monitor their historical earnings to track the overall monetization of their data portfolio.

3.3.2.2. The Enterprise Data Consumer Journey

The enterprise's journey is optimized for efficiency and trust, enabling businesses to acquire high-fidelity insights without the legal liabilities of handling raw personal data.

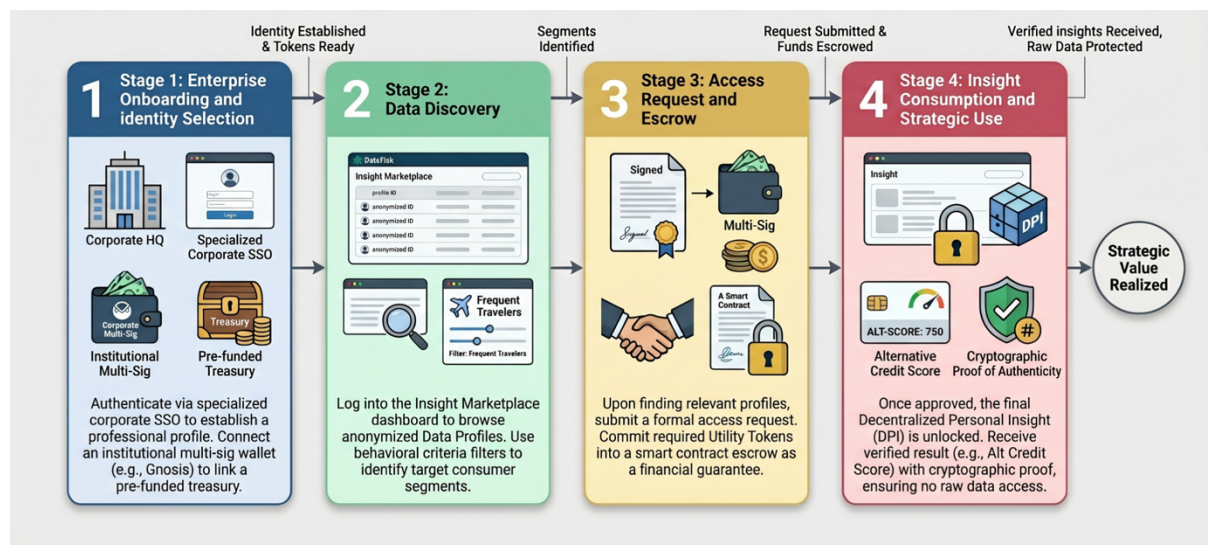


Figure 5 User Journey for Enterprise Data Consumers: From Data Profiles Discovery to DPI Consumptions

In **Stage 1**, the **Enterprise Onboarding and Identity Selection stage**, the enterprise authenticates via a specialized corporate SSO to establish a professional profile. They then connect an institutional wallet (e.g., a corporate MetaMask or Gnosis Multi-Sig) to link their pre-funded treasury to the platform.

In **Stage 2**, the **Data Discovery stage**, the enterprise logs into the Insight Marketplace dashboard to browse anonymized Data Profiles. Rather than viewing personal identities, they use interface filters to query behavioral criteria (e.g., "Frequent Travelers") to identify target consumer segments.

In **Stage 3**, the **Access Request and Escrow stage**, upon finding relevant profiles, the enterprise submits a formal access request. The interface prompts their wallet to commit the required Utility Tokens into a smart contract escrow, serving as a financial guarantee of payment.

In **Stage 4**, the **Insight Consumption and Strategic Use stage**, once the contributor approves the request, the final DPI is unlocked on the enterprise's dashboard. The verified result (e.g., the Alternative Credit Score) is delivered with cryptographic proof of authenticity, ensuring the business receives verified intelligence without ever accessing the underlying raw logs.

By mapping these interconnected journeys, DataTrek transforms the traditionally opaque process of data collection into a transparent, bilateral exchange of value. Ultimately, this user-centric design ensures that the frontend experience remains intuitive, effectively bridging the gap between complex on-chain operations and practical data sovereignty.

3.4 Frontend Interfaces and Feature Integration

Section 3.4 presents the detailed frontend implementation of the DataTrek web application (link: `fite4801.base44.app`). The architecture is bifurcated into two distinct portals - one for Data Contributors and one for Data Consumers - to ensure precise functional separation, while both rely on a shared cryptographic foundation.

3.4.1 Shared Web3 Security Gateway

Both stakeholder portals require users to pass through a unified, decentralized authentication layer before accessing the system's core features.

- **MetaMask Wallet Integration:** The application entirely eliminates centralized username/password databases. Users initiate their session by clicking "Connect Wallet," which triggers a direct interaction with their browser-based MetaMask extension configured to the Sepolia Testnet (see Figure 6).

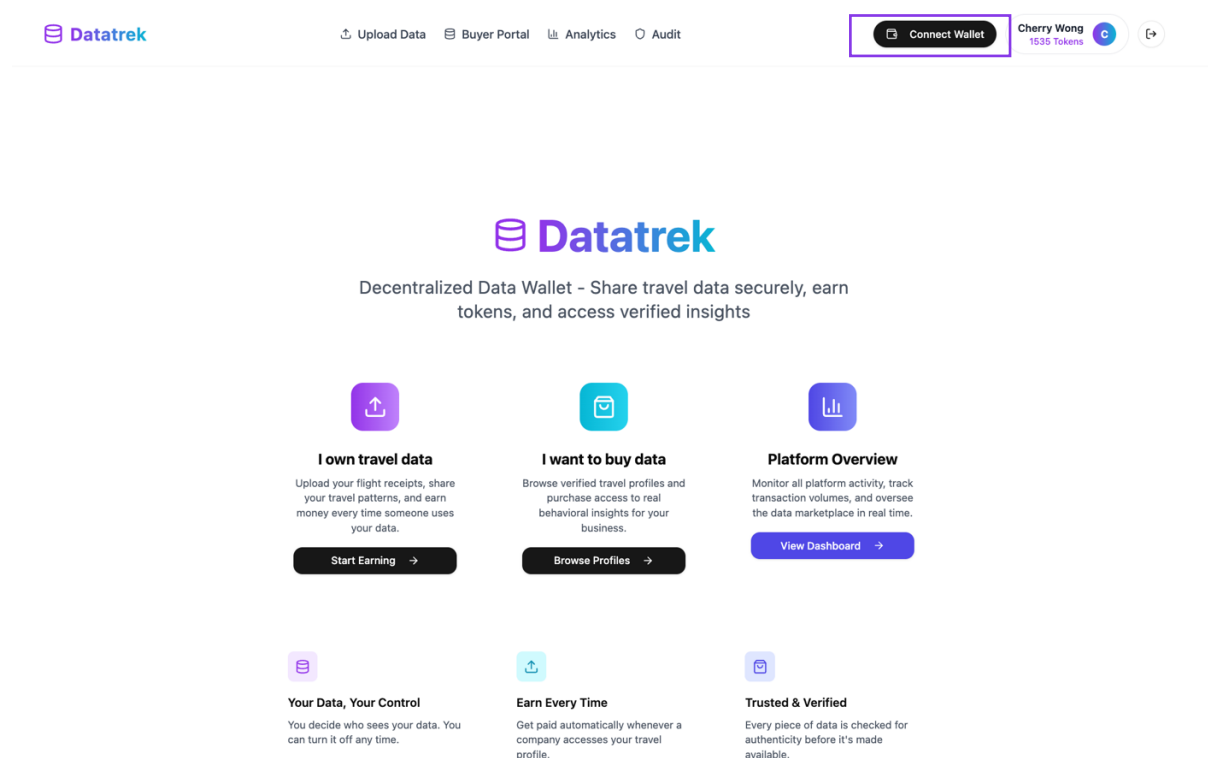


Figure 6 Decentralized Entry Point and MetaMask Extension Integration in DataTrek's HomePage

***Note:** The application replaces traditional username/password databases with a Web3-native "Connect Wallet" trigger. This initiates a direct, secure handshake between the browser-based MetaMask extension and the DataTrek frontend.*

- **Cryptographic Authentication Verification:** To finalize the login, the interface prompts the user to sign a zero-cost cryptographic message. Upon successful signature verification, the frontend state dynamically updates: the "Connect" button is replaced by the user's truncated Ethereum public address and their real-time liquid ETH balance, confirming a secure, authenticated session (see Figure 7 & 8).

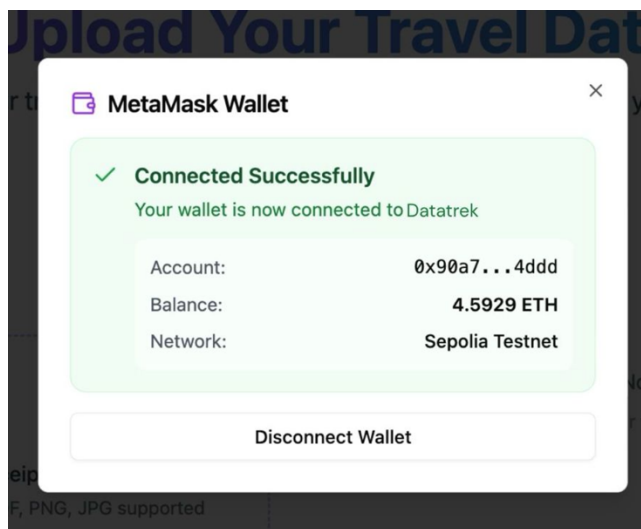


Figure 7 Real-Time Connection Verification and Network Metadata

***Note:** This status window confirms a successful cryptographic link to the Sepolia Testnet. It provides the user with immediate transparency regarding their connected account address and their real-time liquid ETH balance.*

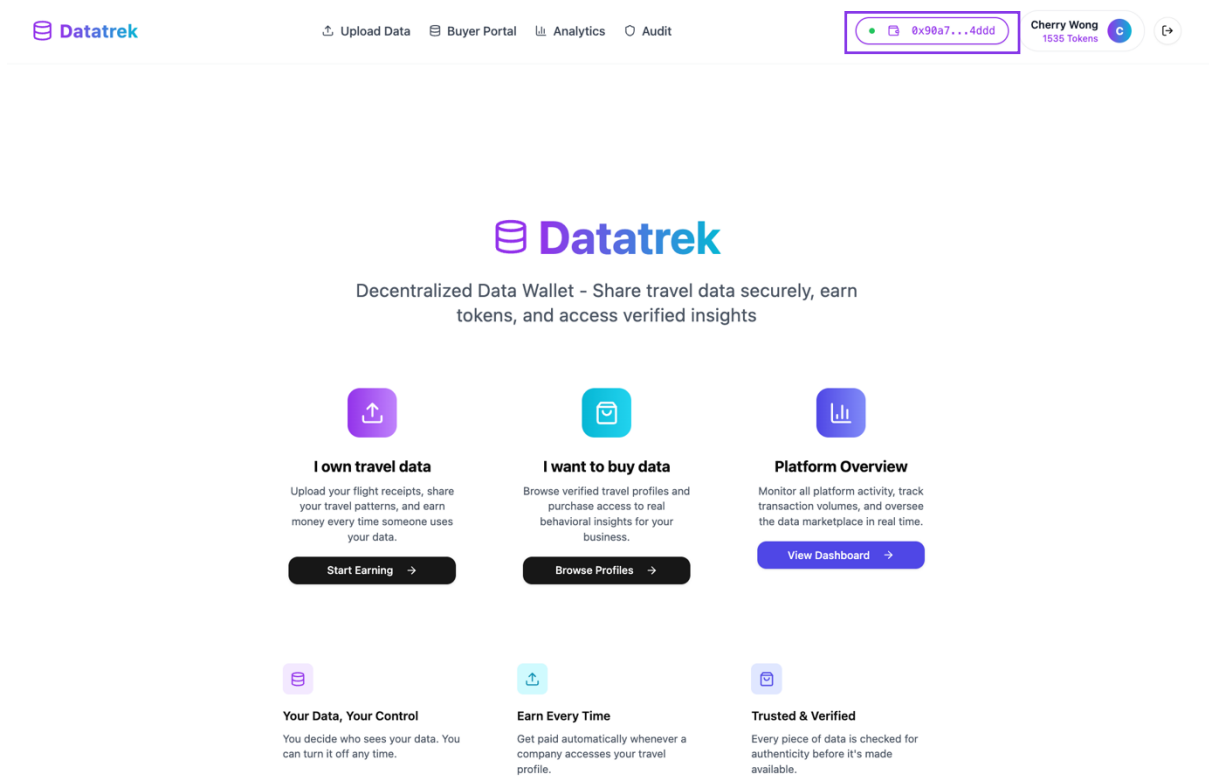


Figure 8 Logged-In Interface and MetaMask Wallet Address Display

***Note:** This view confirms the user is successfully logged in. The original "Connect Wallet" button on the topic right hand corner has been replaced by the user's unique wallet address, showing that the platform now recognizes their decentralized identity.*

3.4.2 The Individual Data Contributor Interface

This portal provides the specific tools required for individuals to ingest raw data, tokenize it, manage privacy settings, and liquidate their earned rewards.

1) Data Ingestion and Valuation Mechanics:

- **Data Upload Module & OCR Scanning:** Contributors utilize a dedicated drag-and-drop zone to submit raw behavioral documents, e.g., PDF travel logs (see Figure 9 & 10). Upon submission, the frontend triggers an off-chain OCR Scanning pipeline to extract structured metadata, e.g., trip dates, destinations, and spend amounts, from the unstructured files.

The screenshot displays the 'Upload Your Travel Data' interface. At the top, a navigation bar includes links for 'Upload Data', 'Buyer Portal', 'Analytics', and 'Audit', alongside a user profile icon labeled 'Lui' with a token balance of '0x44...9827'. The main heading 'Upload Your Travel Data' is centered, with a subtext: 'Share your travel experiences securely and earn tokens. Control your privacy and contribute to better insights.'

The interface is divided into two main sections: 'Upload Travel Data' and 'Data Preview'.

Upload Travel Data: This section features a large dashed box with a cloud upload icon and the text 'Drop your ticket or receipt here. We'll read the data automatically — PDF, PNG, JPG supported'. Below this, there are input fields for 'Full Name *' (e.g., Tom Chan), 'Email Address *' (e.g., tom@example.com), 'Trip Name *' (e.g., Summer in Bali), and a 'Category' dropdown menu (Leisure). A 'Description' text area prompts the user to 'Share details about your trip...'. At the bottom is a prominent 'Upload & Analyze' button.

Data Preview: This section indicates 'No data uploaded yet' and suggests 'Upload your first travel profile to see preview'. Below this, the 'Data Quality & Market Value' section shows a 'Completeness Score' of 0% with the instruction 'Upload more data to improve'. It also displays 'Market Demand' as 'Low' and a 'Travel' data demand of '+0% change'. A 'Suggested Actions' box at the bottom right lists: '+ Add more travel segments to improve completeness' and 'Upload diverse data types for higher rewards'.

Figure 9 Data Contributor's Initial Data Upload Interface and Pre-Data Ingestion Layout

***Note:** This interface presents the standard structure for data ingestion. Users can define a trip, select categories, and drag-and-drop a document (e.g., PDF) into the localized extraction zone to begin the "Data-to-Asset" process.*

 **Upload Travel Data**


 Drop your ticket or receipt here
We'll read the data automatically — PDF, PNG, JPG supported

 CHOW KA HO PROF 30SEP HKG.pdf

0.73 MB

 **Reading your document with AI...**
Extracting flight details, dates, and payment info

Full Name *

Email Address *

Trip Name *

Category

Leisure ▾

Description

Share details about your trip...

Figure 10 Active Document Ingestion and Automated Metadata Parsing

***Note:** Upon detecting the uploaded file (CHOW KA HO PROF. . .pdf), the system invokes an on-page AI service. A real-time status notification informs the user that the backend is actively operationalizing an OCR scanner to read details like dates, flight references, and payment info from the ticket.*

- **IPFS Decentralized Storage:** To enforce strict data sovereignty, the frontend locally encrypts the raw file and transmits it directly to the **IPFS Decentralized Storage** network. The application receives a cryptographic hash (CID) in return, ensuring the raw data remains entirely off-chain.
- **Data Preview Integration:** The interface generates a concise, read-only summary of the extracted behavioral metadata (see Figure 11). This module allows the user to instantly verify the accuracy of the OCR parsing, including trip dates, destinations, and travel categories, ensuring the data is correctly structured before it is processed by the on-chain machine learning engine.
- **Data Quality Review:** Without requiring a page reload, the interface dynamically renders a percentage-based Completeness Score (see Figure 11). This algorithm evaluates the extracted metadata for missing fields and structural formatting, giving the user immediate feedback on their data's viability.

- Market Demand Assessment:** The frontend simultaneously queries active smart contract purchase requests to evaluate the uploaded dataset against current enterprise needs, displaying a real-time desirability indicator (e.g., "High Demand") (see Figure 11).

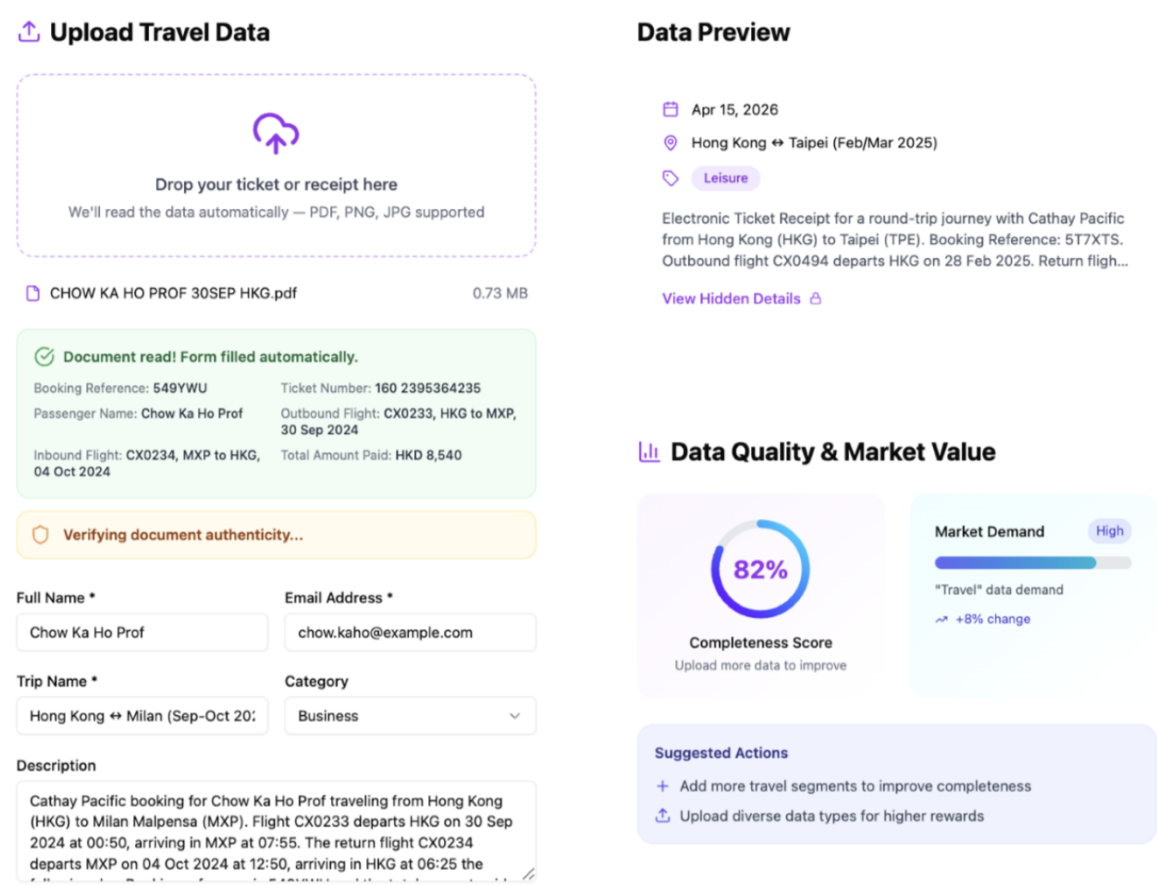


Figure 11 Data Valuation Engine Module: Real-time Analytics on Data Quality and Market Value

***Note:** This integrated view consolidates the final output of the valuation pipeline. It demonstrates the successful automated field population, a sanitized data preview for user verification, a percentage-based Completeness Score, and a real-time Market Demand indicator derived from active enterprise smart contract requests.*

- Data Ownership Tokenization:** Once the upload and validation sequence concludes, the frontend triggers a smart contract execution to mint a SBT. The user's dashboard updates to display this "Digital Deed," anchoring permanent ownership of the data to their MetaMask wallet.

2) Privacy Management and Reward Liquidation:

- **Consent and Anonymous Access Setting:** Contributors navigate to a privacy dashboard where they utilize toggle switches to dictate data visibility. By default, raw identifiable details are masked, ensuring only anonymized metadata is queryable by the market.

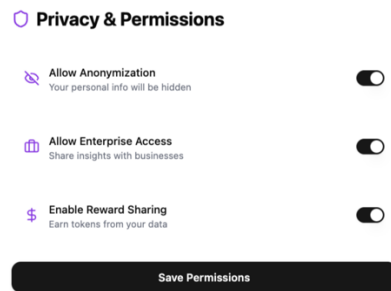


Figure 12 Privacy Governance and Data Access Permissions Toggles

***Note:** This interface allows data contributors to exercise granular sovereignty over their assets by proactively toggling anonymization, enterprise visibility, and reward sharing preferences prior to marketplace listing.*

- **New Purchase Request Notification:** When an enterprise initiates a purchase, the contributor's dashboard surfaces a real-time pop-up notification detailing the buyer's requested use-case and offered token reward, accompanied by explicit "Accept" or "Reject" governance buttons.

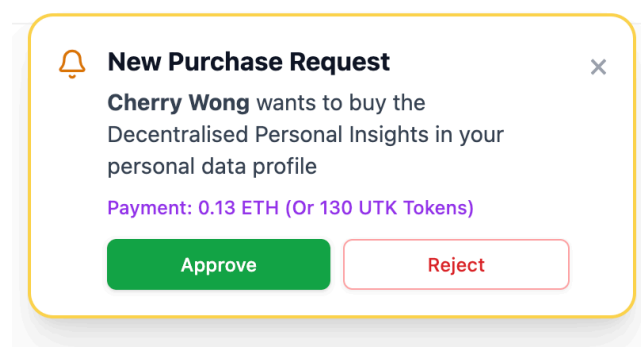


Figure 13 Real-Time Enterprise Purchase Request Pop-up and Consent Verification

***Note:** This notification alerts the contributor to an active enterprise bid, enforcing explicit user governance by requiring manual approval or rejection of the requested data access and corresponding token payment.*

- **Token Reward Dashboard:** A financial interface visualizes the contributor's cumulative platform Utility Tokens earned from approved enterprise requests, displaying their equivalent value in ETH.

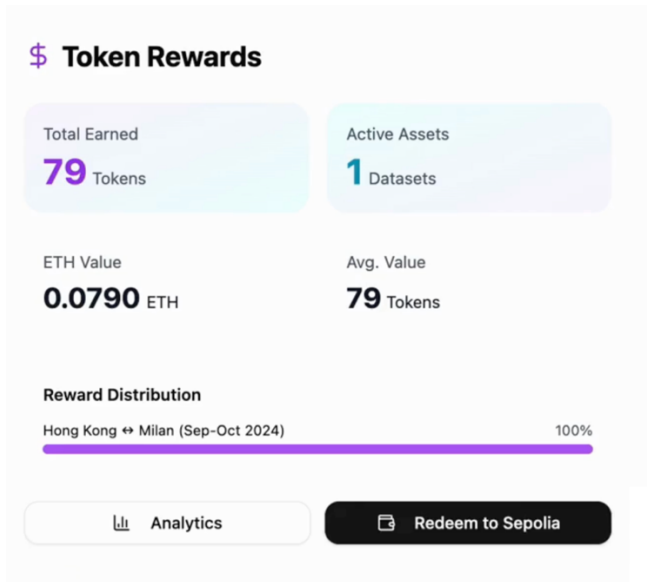


Figure 14 Token Rewards Dashboard and Asset Liquidation Module

***Note:** This financial hub tracks the contributor's cumulative utility token earnings and their real-time ETH equivalent. It serves as the primary gateway for users to initiate the smart contract function that redeems protocol rewards to their connected MetaMask wallet.*

3.4.2 The Enterprise Data Consumer Interface

The initial phase allows enterprises to evaluate and purchase access to behavioral data without compromising individual user privacy.

1) Data Profiles Marketplace Discovery and Trustless Acquisition:

- **Buyer Portal & Marketplace Discovery:** Enterprise users are presented with an aggregated dashboard displaying macro-network statistics (e.g., total available profiles, average network scores, and total network spend) as shown in Figure 14. Below this, a grid of "User Cards" allows buyers to evaluate profiles based on public behavioral metadata (e.g., total trips and average spend). Crucially, to protect the core value of the data asset, the predictive machine learning outputs, specifically the Alternative Credit Score and Health Index, remain cryptographically locked behind a smart contract payroll prior to purchase.

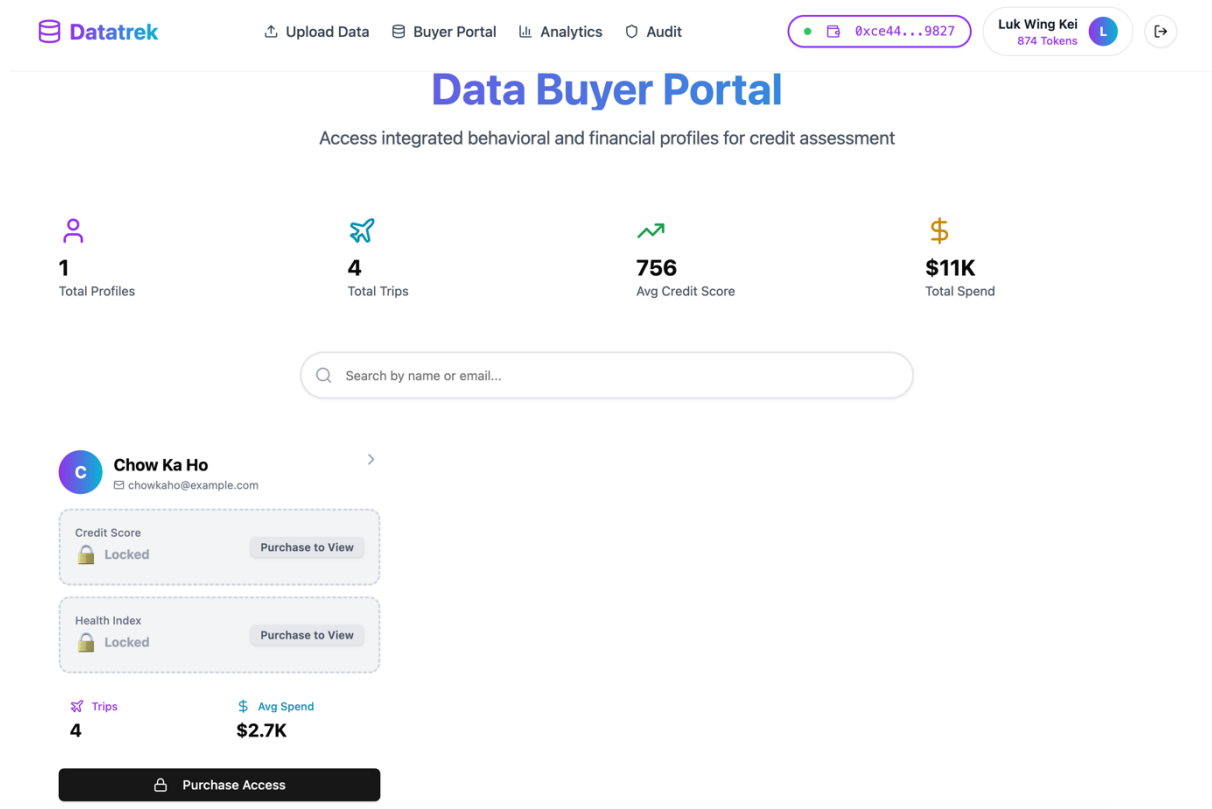


Figure 14 Data Consumer Portal and Marketplace Discovery Dashboard

***Note:** The primary enterprise interface displaying aggregated macro-network statistics (e.g., total profiles, average credit score). Below, individual User Cards present public behavioral metadata (e.g., total trips, average*

spend) to aid buyer evaluation, while keeping the core predictive ML insights (Credit Score and Health Index) strictly locked prior to purchase.

- **Smart Contract Verified Access Requests:** Upon identifying a target profile, the enterprise clicks "Purchase Access" to trigger the transaction modal (see Figure 15). To create a permanent, auditable record of the data's intended use, the consumer must input a specific "Business Use Case." The UI then calculates the total data cost plus the platform fee, allowing the enterprise to review the financial breakdown before cryptographically signing the transaction via MetaMask.

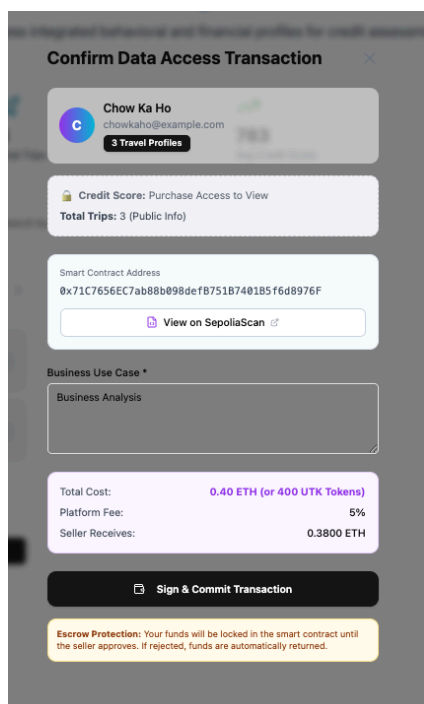


Figure 15 Data Access Confirmation and Financial Breakdown Interface Upon Clicking Purchase Access to a Data Contributor's Data Profile

Note: The primary transaction modal where the enterprise defines their business use case, reviews platform fees, and initiates the cryptographic commitment of utility tokens.

- Sepolia Testnet Explorer Integration:** To guarantee transactional transparency prior to committing funds, the purchase modal includes a "View on SepoliaScan" button (see Figure 16). This dynamic linking feature routes the enterprise directly to the public blockchain explorer, allowing them to audit the specific smart contract source code, verify the execution rules, and review network gas fees on the public ledger.

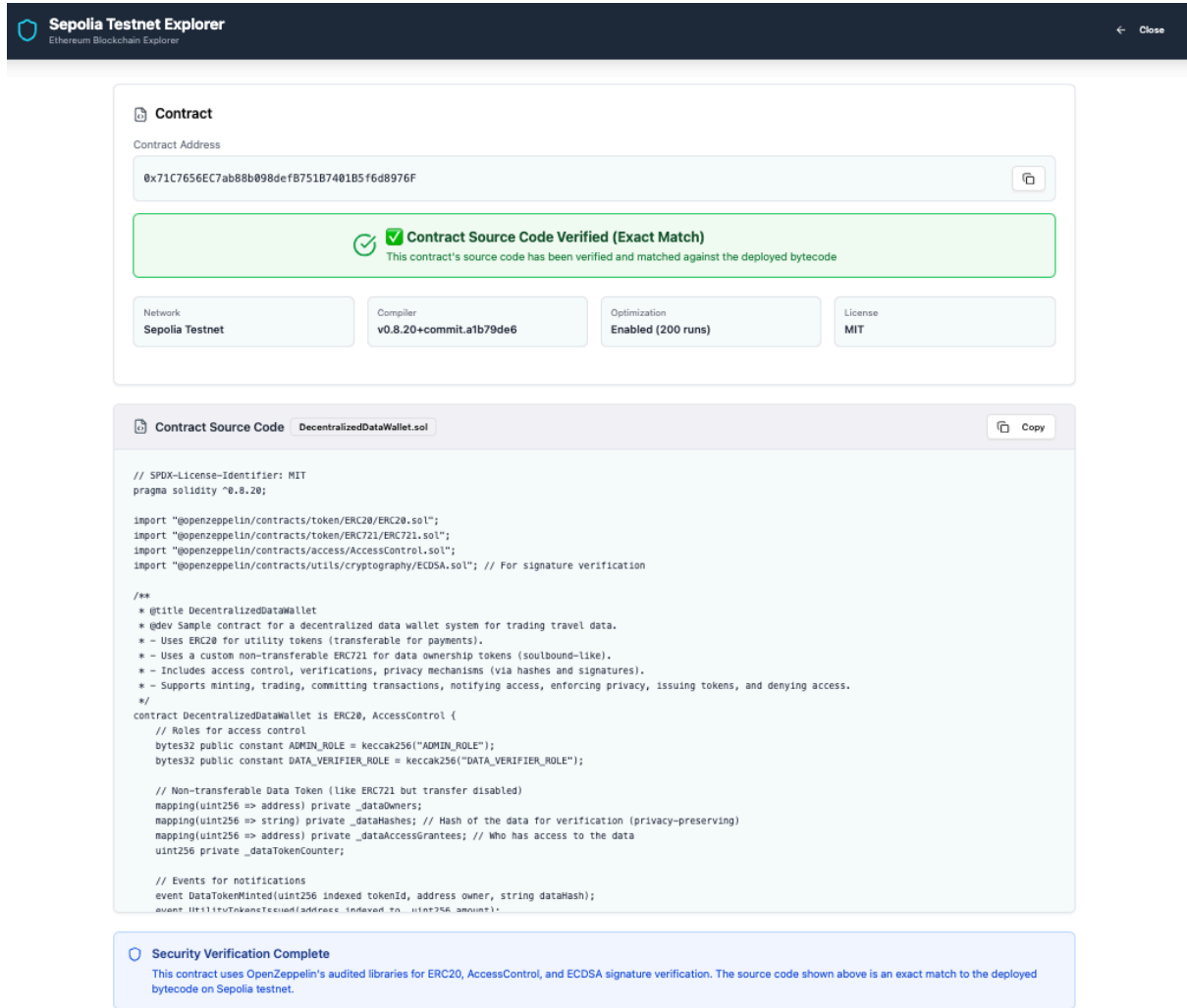


Figure 16 Public Blockchain Audit via Sepolia Testnet Explorer

***Note:** The integrated SepoliaScan interface allows enterprises to independently verify the smart contract's audited source code and execution logic on the public ledger before committing*

- **Escrow Protection and Bidirectional Consent:** Once the transaction is signed, the requested Utility Tokens are locked into a secure smart contract escrow (see Figure 17). The interface enters a pending state ("Waiting for Seller Approval"), enforcing bidirectional consent by holding the enterprise's funds safely until the data contributor explicitly accepts the request on their personal dashboard.

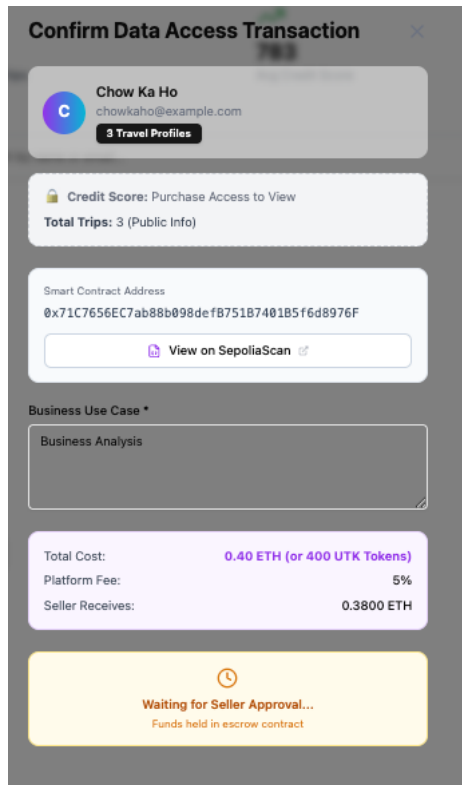


Figure 17 Smart Contract Escrow and Pending Approval State

***Note:** Demonstrates the platform's bidirectional consent mechanism. Enterprise funds are securely locked in the smart contract escrow until the individual data owner grants explicit approval.*

2) On-Chain Computation and Insight Rendering

The final phase occurs immediately following the contributor's approval, transitioning the locked data into actionable business intelligence.

- **Smart Contract Execution & DPI Generation:** Once the Data Contributor clicks "Accept", the smart contract settles the escrowed tokens and triggers the native, on-chain Decision Tree. The contract evaluates the user's localized metadata entirely on-chain, ensuring the raw, identifiable files are never exposed to the enterprise.
- **Insight Rendering (Alternative Credit Score & Health Index):** The interface receives the deterministic mathematical outputs from the on-chain Decision Tree. Without requiring a page reload, the dashboard dynamically updates, removing the "Locked" icons on the User Card to reveal the finalized Alternative Credit Score (scaled 300–850) and the Health Index (scaled 0–100) (see Figure 18).
- **Risk Categorization and Multi-Dimensional Reporting:** To translate the raw on-chain integers into strategic value, the frontend parses the metrics into a comprehensive DPIs report (see Figure 18), which is “Behavioral and Health Insights Report” in this case demonstration. This report automatically applies Risk Categorization and Leveling, grouping the user's behavior into distinct, actionable enterprise tiers, e.g., "High Spending Power," "Low Risk," "Stable Lifestyle", to finalize the value delivery process.

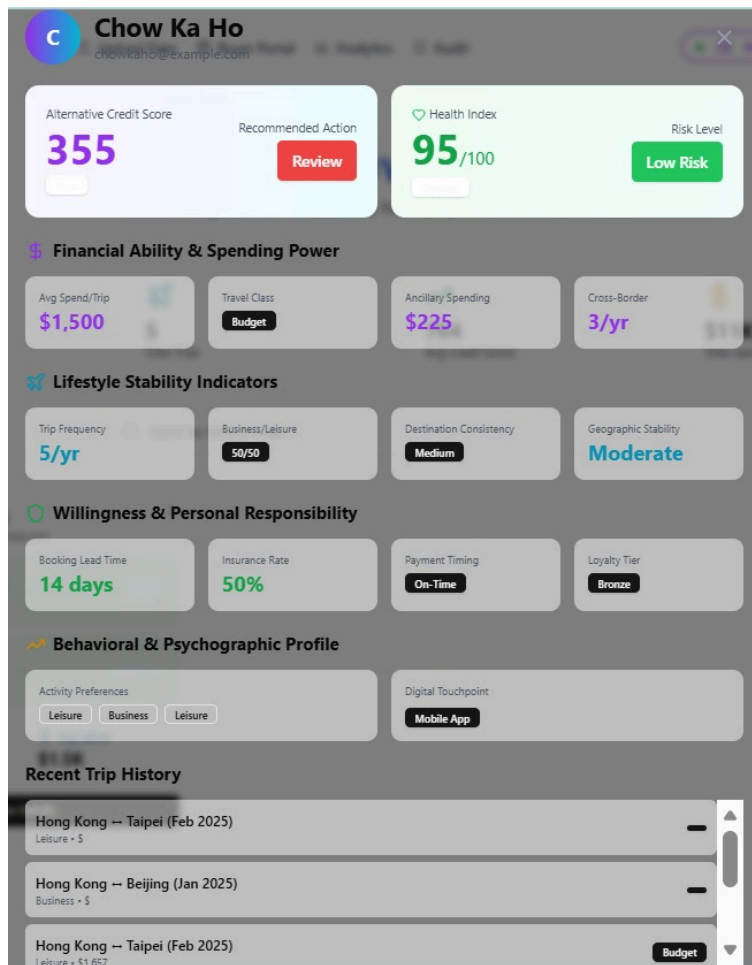


Figure 18 Unlocked Comprehensive Behavioral and Health Insights Report after Successful Purchase of DPI from Individual Data Contributor's Profile

***Note:** The final output of the decentralized exchange, revealing the contributor's deterministic Alternative Credit Score and Health Index generated by the on-chain machine learning models.*

3.4.4 Data Dashboard Analytics for Visualization of Data Monetization

To materialize the platform's core value proposition - shifting data ownership from centralized monopolies to the individual - the frontend features a high-fidelity Data Earnings Dashboard to visualize how their data are monetised (see Figure 19). This interface serves as a real-time analytics hub that visualizes the direct financial yield generated by the user's data assets. By aggregating on-chain payout events from the Web3 wallet with off-chain valuation metadata, the dashboard provides a granular breakdown of earnings per data category, transforming abstract data points into a transparent, liquid asset class.

Figure 19 illustrates the analytics dashboard shown after the DPI transaction. It comprises several key functional modules designed to foster transparency and user engagement:

- **Value Realization Banner:** To psychologically anchor the transition from Web2 to Web3, the hero banner explicitly contrasts the user's previous data earnings without the platform ("0.00") against their newly accrued fiat-equivalent value (e.g., "\$5.81 HKD est."). This real-time fiat conversion helps non-technical users comprehend the tangible value of their utility tokens.
- **Aggregated Network Metrics:** A grid of status cards provides a high-level summary of the user's marketplace participation. This includes the Total Transactions (e.g., 34 times data was accessed), the number of Data Profiles Uploaded (e.g., 4 active on the marketplace), and the total Points Earned (e.g., 278). These metrics represent the exact frequency and compensation volume generated by enterprise smart contracts querying the user's insights.
- **Behavioral Analytics (Your Trip Categories):** The interface parses the user's uploaded metadata to provide a visual breakdown of their data portfolio (e.g., distinguishing between 75.0% Leisure and 25.0% Business travel). This gives the contributor insight into the specific behavioral patterns they are supplying to the ML engine.
- **On-Chain Activity Ledger (Recent Sales Activity):** The "Recent Sales Activity" feed serves as a frontend translation of the blockchain's transaction history. It provides a timestamped, live-updating ledger of every time an enterprise accesses the user's data,

alongside the specific token micro-rewards (e.g., "+162 pts" or "+185 pts") deposited into their smart contract escrow.

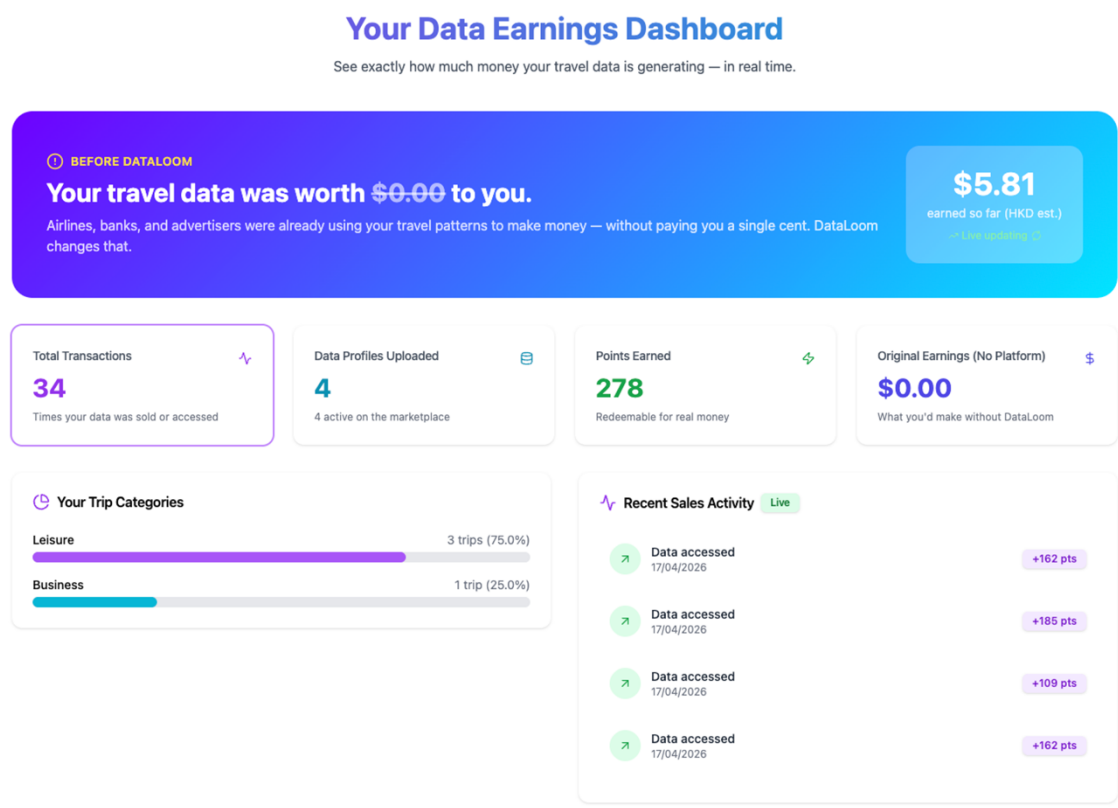


Figure 19 Dashboard: Real-Time Data Monetization and Multi-Dimensional Behavioral Analytics

***Note:** A centralized hub integrating real-time on-chain transaction events with off-chain metadata to visualize financial yields, data upload metrics, and categorical portfolio insights.*

3.5 Native On-Chain Machine Learning Backend Implementation

Section 3.5 describes the backend implementation of the DPIs Generation Engine. This engine transforms verified behavioral data into actionable intelligence. While the initial prototype relied on off-chain execution, the finalized architecture successfully migrates this computational logic natively onto the blockchain. This section details the algorithmic pivot, the feature engineering pipeline, the mathematical synthesis of the Alternative Credit Score and Health Index, and the final backtesting validation.

3.5.1 Comparative ML Algorithmic Models

The DPI Generation Engine utilizes a sophisticated machine learning framework to transform heterogeneous behavioral data into verifiable risk metrics. As established in the interim development phase, multiple architectural alternatives were rigorously evaluated to address the non-linear nature of behavioral data.

Initially, linear and logistic regression models were excluded because they rely on first-order dependencies and cannot capture complex multivariate interactions. K-Nearest Neighbors (KNN) was disqualified due to high computational latency and its sensitivity to the "curse of dimensionality." Neural networks, despite their predictive power, were dismissed owing to their "black box" nature. Consequently, during the interim phase, a 43-tree Random Forest Classifier was selected as the optimal ensemble learner for off-chain execution, as utilizing ensemble architectures to reduce variance is a standard industry practice for credit risk modelling [21].

3.5.2 Algorithmic Evolution and Web3 Constraints

However, transitioning the platform from an off-chain server toward **native on-chain execution** introduced strict Web3 engineering constraints that made the 43-tree Random Forest model unviable:

1. **Inference Latency and Aggregation Bottlenecks:** The primary drawback of a Random Forest is the latency caused by Bootstrap Aggregating (Bagging). To generate a single credit score, the system must traverse all 43 individual decision trees and compute a majority vote. As attempted on-chain, this heavy computational aggregation introduces unacceptable delays, breaking the synchronous, real-time API polling required by enterprise consumers.
2. **The Gas Ceiling:** Executing the conditional branching logic for a 43-tree ensemble natively on the EVM requires massive computational overhead. This incurs prohibitive gas costs for the end-user and routinely exceeds Ethereum's block gas limits.
3. **EIP-170 Size Limits:** The Ethereum protocol enforces a strict 24.576 KB size limit on compiled smart contracts [32]. The sheer volume of code required to hardcode 43 distinct trees easily breaches this limit, rendering the contract impossible to deploy.

To maintain predictive accuracy while ensuring economic viability and zero-latency execution, the architecture was systematically pivoted. Lightweight, deterministic Decision Tree Classifiers were selected as the optimal architecture for the finalized on-chain engine, providing a feasible path for decentralized verification without exceeding EVM constraints or introducing aggregation latency.

3.5.3 Data Ingestion and Feature Engineering Pipeline

Before detailing the data ingestion pipeline, it is necessary to contextualize the chosen ML models. The system relies on two primary data evaluation tracks: a Credit Oracle and a Health Oracle. These two specific application scenarios were selected because they represent highly monopolized, high-value data sectors within FinTech and InsurTech. Demonstrating DataTrek's capabilities in these fields effectively proves the platform's commercial viability to enterprise consumers while highlighting the maximum potential tokenized value for individual data contributors.

A robust feature engineering pipeline was implemented in the Python backend (`server.py`) to bridge the gap between raw off-chain data and the EVM. Because the EVM natively operates on `uint256` integers and lacks support for complex text strings, the pipeline acts as a pre-processing layer to standardize and scale inputs.

The system extracts metadata from user-uploaded PDFs via off-chain OCR and applies heuristic rules to generate engineered features. This transformation ensures the decision trees can execute cost-effectively on-chain by minimizing computational overhead.

Raw Data Field	Engineered Feature	Transformation Logic	Feature Proxy	System Rationale & Purpose
Annual Income	<code>creditAmount</code>	$\text{Credit Amount} = \frac{\text{Annual Income}}{100}$	Spending Proxy	Scales down fiat values into standardized metrics to reduce EVM gas consumption.
Frequent Flyer	<code>tripsAnnual</code>	"Yes" → 18; "No" → 2	Flight Frequency Proxy	Converts categorical text into discrete numeric values to evaluate lifestyle stability.
Profile Quality	<code>duration</code>	$\text{Score} \times 0.6$	Account Maturity Proxy	Simulates account longevity within the financial system.
Trip Frequency	<code>installment</code>	< 5(1), < 10(2), < 15(3), ≥ 15(4)	Repayment Capacity	Buckets travel frequency to estimate reliable installment capacity.
Birthdate	<code>age</code>	Extracted as Integer	Life-Stage Risk	Directly parsed as a primary splitting node in both decision trees.

Table 10 Data Mapping and Feature Transformation Logic

3.5.4 Off-Chain Model Training and Automated Pruning

The backend operationalizes the "Data-to-Asset" pipeline through a structured sequence of model training and real-time feature engineering. Before deploying the deterministic rules to the blockchain, the models were trained off-chain using the Python-based `scikit-learn` ML framework [33]. Two distinct open-source datasets were utilized for training and backtesting: the *German Credit Dataset* (comprising **1,000** financial records) for the Credit Oracle [34], and the *Travel Insurance Dataset* (comprising **1,987** behavioral profiles) for the Health Oracle [35]. Both datasets are publicly accessible via Kaggle to ensure reproducibility.

The raw travel behavioral metadata, extracted from user-uploaded PDFs via off-chain OCR, required rigorous standardization. Unstructured data points were engineered into discrete, quantifiable variables capable of on-chain ingestion [22]:

- **Credit Oracle Features:** Age, Credit Amount (derived as Annual Income/100), Duration (account age from financial logs), Installment ratios, and Residence length.
- **Health Oracle Features:** Age, Credit Amount, TripsAnnual (mapped from Frequent Flyer status; e.g., 'Yes' = 18, 'No' = 2), and Insurance Rate percentages.

During the training phase, the algorithm utilized Recursive Binary Splitting, finding splits that minimized the **Gini Impurity**:

$$Gini(D) = 1 - \sum_{i=1}^c p_i^2$$

Where:

- D represents the specific dataset or node currently being evaluated.
- c is the total number of target classes (e.g., binary outcomes such as "Purchased" vs. "No Insurance").
- p_i is the probability (or relative frequency) of an item belonging to class i within that specific node.

A critical architectural success occurred during the Health Oracle training through the process of automated feature pruning. While four features were initially passed into the `scikit-learn`

model, the algorithm was strictly constrained to a was constrained to a maximum depth of 2 (Health) and 3 (Alternative Credit) to optimize for Ethereum gas limits.

During the Health Oracle training, the Decision Trees algorithm mathematically determined that `creditAmount` (Income) and age provided the highest Information Gain. Consequently, the model pruned the `tripsAnnual` feature entirely, selecting Income as the root node and Age as the secondary split. This ensured that only the most statistically significant predictors were prioritized for on-chain execution, maximizing model utility while maintaining a predictable $O(d)$ gas profile.

3.5.5 Transpilation and Floating-Point Arithmetic Scaling

Deploying the trained models to the EVM required complex architectural adaptations, primarily because the Solidity programming language and the EVM natively lack support for floating-point mathematics (decimals) and standard ML libraries.

In traditional Python-based ML environments, a model might return a continuous probability like 0.85. Because Solidity can only process whole integers, transferring this logic natively on-chain required a custom transpilation script (`tree_to_solidity`). This script extracts the nodes and terminal leaves from the `scikit-learn` model and automatically generates nested `if/else` conditional logic.

To solve the floating-point limitation, an **integer arithmetic scaling mechanism** was employed [36]. Fractional probabilities are scaled up and stored as whole integers within the smart contract (e.g., 0.85 is natively stored as 85). To ensure the contracts remained within EVM gas and size limits, strict pruning was enforced:

- **The Credit Oracle:** Constrained to a maximum depth of 3. This strict boundary ensures that the algorithm generates no more than 8 terminal leaves (2^3). During on-chain execution, a user's data will pass through a maximum of three `if/else` conditional checks before reaching a final probability.
- **The Health Oracle:** Constrained to an even tighter maximum depth of 2, generating a maximum of 4 terminal leaves (2^2). This ensures that the health insight evaluation requires a maximum of only two computational steps, further minimizing network latency and enterprise transaction fees.

By bounding the time complexity to $O(d)$, where d is the tree depth, the smart contract guarantees a highly predictable worst-case scenario for gas consumption.

3.5.6 DPI Synthesis: Credit Scoring and Health Index Stratification

During the real-time inference stage, the transpiled smart contracts generate discrete behavioral outcomes. To make these mathematically viable for enterprise institutions, precise integer normalizations are applied on-chain to map these outputs to standard industry ranges.

1. Alternative Credit Score Synthesis

Before calculating the final FICO-style score, the smart contract must first determine the base Probability of the user being a reliable borrower. This integer value is derived directly from the transpiled Decision Tree executing on-chain. As the user's metadata (specifically `duration`, `creditAmount`, and `age`) is processed through the nested `if/else` conditional branches of the smart contract, it eventually reaches a terminal leaf node. Each leaf node holds a hardcoded integer (ranging from 10 to 95) representing the statistical confidence of a positive repayment outcome based on the off-chain training data.

Once the contract assigns this specific Probability (for example, 85), it is plugged into the following integer scaling formula. This maps the probability to the standard 300–850 industry range established by the FICO Score [23]:

$$Credit_Score = 300 + \left[\frac{Probability \times 550}{100} \right]$$

By multiplying the integer probability by the 550-point spread before dividing by 100, the smart contract successfully prevents integer truncation errors and bypasses the EVM's inability to calculate floating-point decimals.

Finally, a deterministic risk stratification function converts this finalized numerical score into predefined risk tiers directly inside the smart contract, emitting the exact rating required to operationalize the score within existing automated enterprise underwriting pipelines and risk-control frameworks [24] (see Table 5):

Risk Tier (Rating)	Score Range	On-Chain Logic Threshold	Behavioral and Financial Implication
Excellent	740 - 850	<code>score >= 740</code>	Exceptional financial capacity and superior behavioral consistency.
Good	670 - 739	<code>score >= 670</code>	Moderate reliability and stable, consistent behavioral patterns.
Fair	580 - 669	<code>score >= 580</code>	Average reliability, may exhibit minor inconsistencies in data.
Poor	300 - 579	<code>else fallback</code>	High risk, signifies significant volatility or low transparency.

Table 5 On-Chain Alternative Credit Risk Stratification Tiers

Note: While the interim prototype utilized a 5-tier stratification, the finalized on-chain architecture merges the top percentiles into a consolidated 'Excellent' tier (≥ 740) to minimize conditional branching and optimize gas consumption within the EVM.

2. Health & Wellbeing Index Synthesis

Unlike the credit score, the Health Index does not use a continuous mathematical multiplier. Because the on-chain model was heavily restricted to a depth-2 Decision Tree for extreme gas optimization, the algorithm assigns discrete, hardcoded index values based on the terminal leaf nodes of the tree.

The exact transpiled Decision Tree logic executing in the Solidity contract (`calculateHealthInsight`) is represented by the following piecewise function, evaluating `creditAmount` (c) and `Age` (a):

$$Health\ Index\ (c, a) = \begin{cases} 45 & \text{if } c \leq 13000 \text{ and } a \leq 31 \\ 65 & \text{if } c \leq 13000 \text{ and } a > 31 \\ 85 & \text{if } c > 13000 \text{ and } a \leq 28 \\ 98 & \text{if } c > 13000 \text{ and } a > 28 \end{cases}$$

This algorithmic structure accurately stratifies contributors into four highly specific health and lifestyle risk categories. This provides targeted, deterministic insights for health and insurance enterprise consumers based purely on objective, on-chain conditions (see Table 6):

Insight Status	Base Health Score	Smart Contract Trigger Conditions	Lifestyle Implication
Highly Optimal	98	High Spend (> 13000), Older Demographic (> 28)	Superior travel preparedness and highly stable lifestyle.
Optimal	85	High Spend (> 13000), Young Demographic (≤ 28)	High preparedness, active lifestyle, consistent compliance.
Stable	65	Low Spend (≤ 13000), Older Demographic (> 31)	Average health stability, occasional gaps in risk management.
Vulnerable	45	Low Spend (≤ 13000), Young Demographic (≤ 31)	High Risk: Low travel preparedness and chronic risk exposure.

Table 6 On-Chain Health & Wellbeing Index Stratification

***Note:** By executing this piecewise logic natively on the EVM, the protocol guarantees that health risk assessments remain completely deterministic and transparent. This allows enterprise consumers to acquire standardized risk profiles while fully abstracting and protecting the contributor's raw personal metadata.*

3.5.7 API Gateway and Secure Data Decryption

To facilitate seamless integration for enterprise systems, the platform utilizes a secure Web3 API Gateway to bridge off-chain queries with on-chain execution. The insight retrieval process follows a strict, three-stage cryptographic workflow:

- **API Gateway Authentication:** As shown in Figure 20, the process begins at the Private Data Portal. Enterprise consumers must input the specific Data Owner's Name alongside a cryptographically verified API Key. This ensures that only authorized buyers who have successfully completed the escrow and consent phases can initiate queries against the individual's dataset.

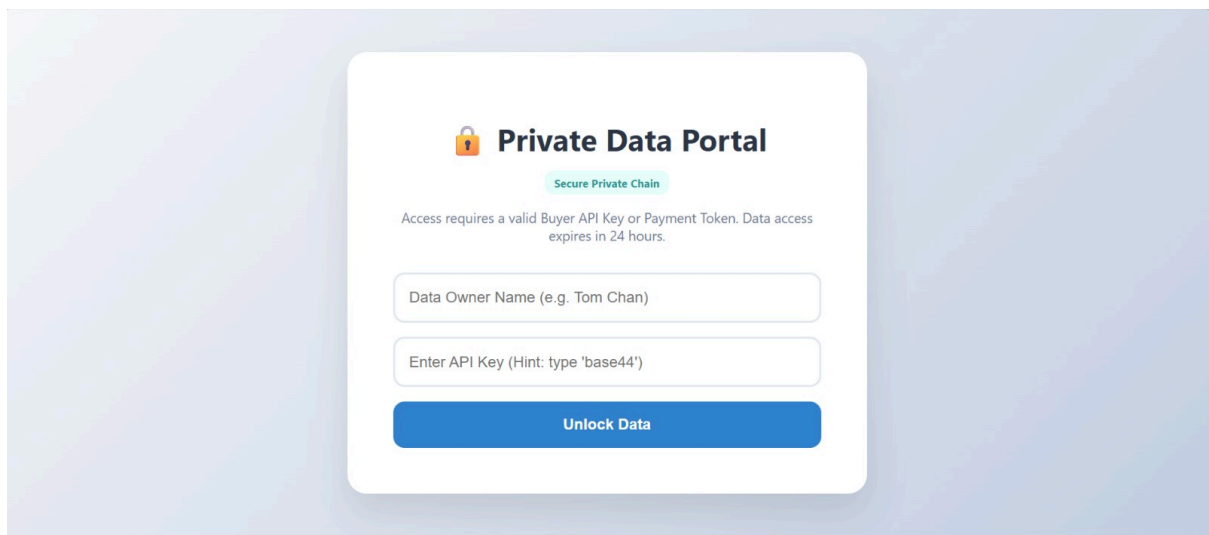
The image shows a web interface for a 'Private Data Portal'. At the top, there is a lock icon followed by the title 'Private Data Portal'. Below the title is a green badge that says 'Secure Private Chain'. A line of text states: 'Access requires a valid Buyer API Key or Payment Token. Data access expires in 24 hours.' There are two input fields: the first is labeled 'Data Owner Name (e.g. Tom Chan)' and the second is labeled 'Enter API Key (Hint: type 'base44')'. Below these fields is a prominent blue button labeled 'Unlock Data'.

Figure 20 Data Consumer's API Gateway Interface for Secure Insights Retrieval for Enterprises

Note: The enterprise API gateway where data consumers can initiate asynchronous API calls to interact with the on-chain ML smart contracts.

- On-Chain Computation Execution:** Once authenticated, the enterprise accesses the insight generation dashboard (Figure 21). Rather than processing data on a centralized backend, the UI allows the enterprise to manually trigger the native on-chain Decision Tree smart contracts. Clicking to run the Credit or Health algorithms prompts a MetaMask transaction request, requiring the enterprise to authorize the network gas fees necessary to execute the machine learning computation directly on the EVM.

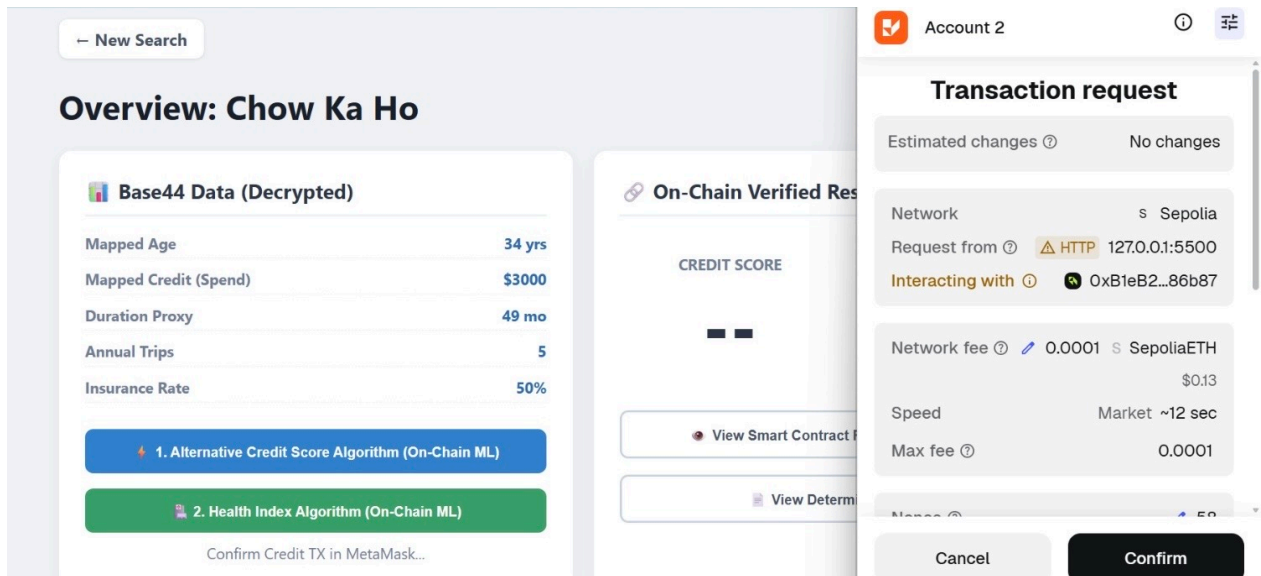


Figure 21 On-Chain Computation Trigger and MetaMask Execution Verification

Note: The enterprise dashboard interface where the buyer triggers the specific smart contract algorithms (Credit and Health). This prompts a MetaMask signature request to execute the machine learning models natively on-chain.

- Event Listening and Insight Rendering:** Crucially, the raw behavioral metadata is never exposed to the enterprise during this process. Instead, the application listens for the specific `ScoreCalculated` and `HealthCalculated` events emitted by the Solidity contract once the transaction is mined. As illustrated in Figure 22, the frontend securely fetches these execution results and renders the finalized deterministic integers—the Alternative Credit Score (e.g., 355) and the Health Index (e.g., 45/100).

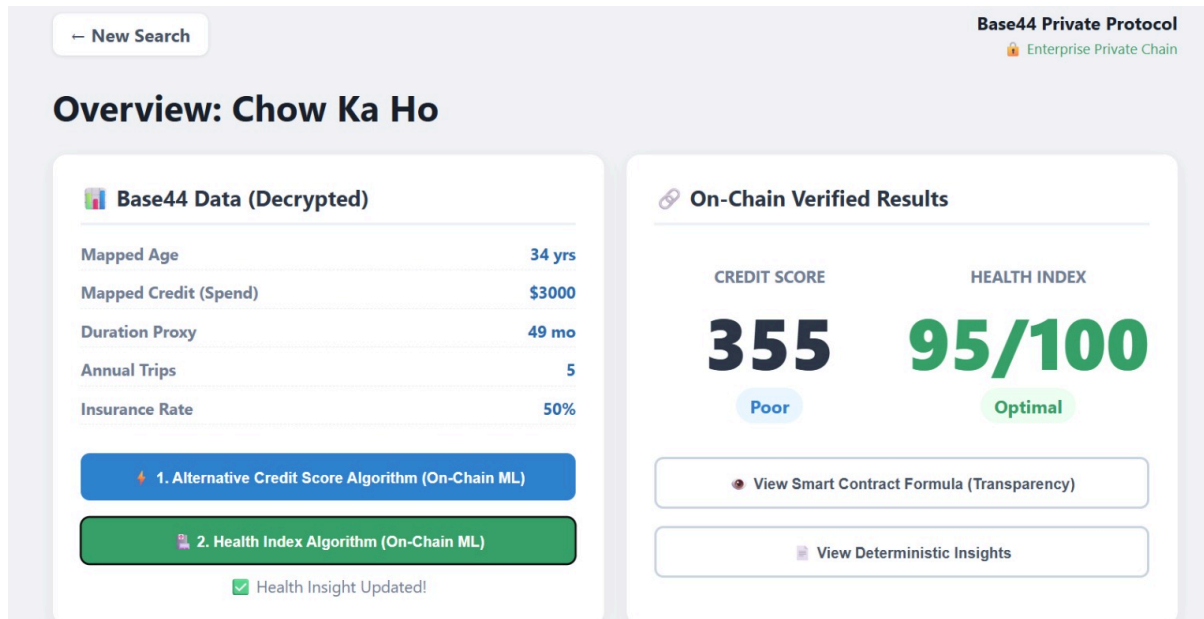


Figure 22 Final Rendered On-Chain ML Insights Dashboard of a Data Contributor's DPI

***Note:** The final UI state displaying the dynamically retrieved on-chain verification results. The raw behavioral metadata remains hidden, while the processed Credit Score and Health Index are delivered to the enterprise.*

3.5.8 Model Backtesting and Performance Validation

Prior to immutable deployment on the Sepolia testnet, both transpiled models underwent rigorous off-chain backtesting to validate predictive accuracy and algorithmic stability.

3.5.8.1 Credit Oracle Validation

The Credit Oracle achieved a baseline accuracy of **70.50%**, as tested against the German Credit Dataset [34] (link: <https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data>). Crucially, the model demonstrated a highly effective 94% recall rate in accurately identifying "Good Borrowers," ensuring minimal missed opportunities for lenders (see Table 7).

Metric	Bad Borrower	Good Borrower	Macro Avg
Precision	0.50	0.73	0.61
Recall	0.15	0.94	0.54
F1-Score	0.23	0.82	0.53
Support	59	141	Total: 200

Table 7 Credit Oracle Backtesting Metrics

Confusion Matrix Output:

- **True Positives** (Predicted Good, Actually Good): 132
- **True Negatives** (Predicted Bad, Actually Bad): 9
- **False Positives** (Predicted Good, Actually Bad): 50
- **False Negatives** (Predicted Bad, Actually Good): 9

***Note:** The high recall for Good Borrowers (0.94) indicates the model reliably identifies low-risk profiles, though the lower precision for Bad Borrowers reflects the necessary predictive trade-off incurred by constraining the Decision Tree to a depth of three for Ethereum gas optimization.*

3.5.8.2 Health Oracle Validation

The Health Oracle was tested against a split of 1,987 profiles from the Kaggle Travel Insurance Dataset [35] (link: <https://www.kaggle.com/datasets/tejashvi14/travel-insurance-prediction-data>). The model achieved a strong overall accuracy of **81.75%**. Behavioral markers found in travel data—such as travel frequency, age-related travel patterns, and the propensity to purchase comprehensive travel insurance—often serve as strong predictive proxies for an individual's underlying health status and risk aversion. As demonstrated by the dataset's features, individuals with chronic conditions or higher health risks exhibit distinct travel and purchasing behaviors, allowing the algorithm to confidently infer a broader health and stability index.

More importantly, the model proved highly accurate at identifying vulnerable users (Class 0: No Insurance), achieving a 95% Recall rate. This makes the Health Oracle an exceptionally conservative and safe tool for enterprise risk assessment, as it rarely misclassifies high-risk individuals as safe.

Metric	Class 0 (No Insurance)	Class 1 (Purchased)
Precision	0.80	0.87
Recall	0.95	0.58
F1-Score	0.87	0.69

Table 8 Health Oracle Backtesting Metrics

Confusion Matrix Output:

- True Negatives (Predicted No Insurance, Actually No Insurance): 244
- False Positives (Predicted Purchased, Actually No Insurance): 13
- False Negatives (Predicted No Insurance, Actually Purchased): 60
- True Positives (Predicted Purchased, Actually Purchased): 81

***Note:** The exceptional recall rate of 0.95 for Class 0 demonstrates the Health Oracle's strength as a conservative underwriting tool, ensuring that highly vulnerable profiles are almost never mistakenly classified as stable to enterprise consumers.*

This successful dual-validations and backtests confirmed that both models surpassed baseline viability, providing the mathematical confidence to hardcode the intelligence into Solidity and finalize the decentralized execution environment.

3.6 System Evaluation and Testing Results

Following the successful integration of the frontend interfaces and the native on-chain ML backend, a comprehensive system evaluation was conducted. As outlined in Section 2.4, this evaluation transitioned the prototype from a theoretical PoC into a practically deployed, user-centered application.

To provide a rigorous assessment of DataTrek’s performance and deployment readiness, the system was evaluated across five primary technical and operational dimensions: **(1) Mathematical and Extraction Accuracy**, validating the integrity of the OCR and ML pipelines; **(2) Computational Efficiency**, measuring the $O(d)$ complexity and gas optimization of on-chain logic; **(3) System Latency**, quantifying the end-to-end round-trip delay; **(4) Economic Viability**, analyzing the micro-transaction feasibility against mainnet gas costs; and **(5) Privacy and Transparency**, assessing the zero-exposure data model and algorithmic explainability. This multi-dimensional framework ensures the prototype is tested not only for functional correctness but for real-world engineering practicality.

3.6.1 Phase I: Alpha Testing Results

The Alpha testing phase focused on internal technical validation and operational assessment. The objective was to evaluate the architectural stability, computational efficiency, and infrastructural resilience of the pure on-chain ML logic prior to user exposure.

The system was heavily stress-tested using simulated data and subjected to real-world network conditions on the Sepolia testnet. While the core mathematical models and cryptographic privacy frameworks functioned flawlessly, evaluating the system under live Web3 constraints exposed severe infrastructural realities. Most notably, the strict rate limits of Sepolia testnet faucets (0.05 SepETH per drip) served as a highly realistic simulation of the treasury management friction a startup faces, while the 14 - 16 second system latency exposed an immediate bottleneck for high-frequency enterprise batch processing.

The complete findings, separated by technical verification and operational logistics, are synthesized below.

3.6.1.1 Technical Evaluation

The technical evaluation focused on the core decentralized logic (see Table 9). The primary goal was to verify the mathematical integrity of the on-chain ML inferences, the efficiency of the smart contracts, and the strict isolation of the cryptographic privacy framework.

System Component	Objective	Observed Outcome, Performance Metric & Proof	Final Status & Engineering Takeaway
Data-to-Asset Pipeline	Validate transformation of raw PDFs into structured metadata.	OCR pipeline successfully processed simulated itineraries and mapped features to the exact integer formats required on-chain.	Passed: Confirmed functional integrity of the off-chain data bridging pipeline.
On-Chain ML Inference	Evaluate computational efficiency of transpiled Decision Trees.	O(d) bounded models proved highly efficient. Execution costs tracked at exactly 46,500 gas (calculateCreditScore) and 44,200 gas (calculateHealthInsight). <i>Note: This is verified via Sepolia Etherscan on deployed contract 0xB1eB...86b87.</i>	Passed: Proved extreme economic viability. At 46,500 gas, the computational cost is negligible, though the economic model remains sensitive to Mainnet gas price volatility.
System Latency	Measure API/Blockchain round-trip execution time.	End-to-end latency tracked at 14-16 seconds . <i>Note: This is verified via console logs, confirmed 2-4 seconds for the Python API fetch + exactly 12 seconds for the Ethers.js await tx.wait() promise to resolve, matching the Ethereum PoS block slot.</i>	Bottleneck Identified: Unviable for high-frequency batch processing; necessitates future L2 scaling.
Privacy Framework	Ensure raw behavioral data isolation.	Verified raw data (actual PDF files) remained completely off-chain; EVM successfully processed only mathematically abstracted variables.	Passed: Confirmed successful implementation of the zero-exposure privacy model.

Table 9 Technical Evaluation Metrics and Outcomes in Alpha Testing

3.6.1.2 Operational Assessment

The operational evaluation focused on the system's resilience when interacting with real-world blockchain infrastructure (see Table 10). This phase intentionally tested for pipeline breaks, gateway failures, and network liquidity constraints to ensure the architecture could survive outside of a local development environment.

Operational Challenge	Root Cause	Observed Impact	Refinement / Mitigation Status
Testnet Liquidity Bottleneck	Strict Sepolia faucet limits (~0.05 SepETH/drip) and multi-day wallet cooldowns.	High-volume simulated queries were logistically impossible to execute consecutively without depleting test funds.	Simulated Reality: Testing schedule adjusted. Served as a realistic simulation of mainnet gas treasury friction.
Off-Chain API Timeouts	Hardcoded network limitations in the Python backend (<code>timeout=10</code> in <code>server.py</code>).	Processing dense PDF files caused the OCR extraction to exceed the 10-second limit, breaking the ingestion pipeline before syncing.	Resolved: Patched the ingestion pipeline with asynchronous processing to decouple UI from backend extraction constraints.
Web3 Gas Estimation	Inaccurate dynamic gas limit prediction during network congestion.	Occasional transaction reversions ("Out of Gas" errors), resulting in failed API calls and loss of limited Sepolia funds.	Resolved: Integrated a dynamic gas limit buffer into the Ethers.js transaction parameters to ensure execution during volatility.
Network Stability	High volatility and congestion on the Sepolia testnet nodes.	Fluctuating transaction processing times and occasional dropped requests during peak congestion.	Resolved: Implemented automated retry logic in the API Gateway to handle pending/stuck transactions.

Table 10 Operational Assessment and Logistical Resilience in Alpha Testing

3.6.2 Phase II: Beta Testing Results

Following the technical stabilization in Phase I, Beta Testing was conducted to assess the prototype's practical utility. As outlined in the methodology (Section 2.4.2), the project deliberately pivoted from a theoretical PoC utilizing synthetic data to a practical deployment evaluating real-world data applications.

To maximize qualitative depth and facilitate rapid iteration, the testing pool was restricted to two distinct participants with contrasting technical backgrounds. Crucially, each participant was tasked with testing the platform from two perspectives: first acting as the Data Contributor (uploading their own travel receipts), and subsequently acting as the Data Consumer (acting as an enterprise underwriter querying the system). The evaluation utilized an unstructured "Think-Aloud" observational protocol to capture genuine UI/UX friction points in a naturalistic setting.

Participant Profiles:

3. **Participant 1 (The Web3 Enthusiast):** A 21-year-old HKU undergraduate, majoring in Economic and Finance, trusting the Web3 ecosystems.
4. **Participant 2 (The Web3 Skeptic):** A 24-year-old AI Software Engineer currently based in San Francisco, majored in Computer Science and Mathematics. Highly analytical, but skeptical of Web3 applications.

3.6.2.1 Quantitative Evaluation

Rather than expecting flawless execution, the quantitative evaluation tracked the functional completion, extraction accuracy, and system-wide latency of the prototype when handling the unpredictable formatting of real-world airline documents.

To ensure rigorous reporting, system latency was tracked directly via the application architecture: the end-to-end latency was measured using standard frontend performance tracking. In our JavaScript code, we captured a timestamp right before initiating the `Ethers.js` `await tx.wait()` function, and captured a second timestamp the moment the promise resolved and the blockchain receipt was returned. The difference between these timestamps accurately reflects the time spent in the mempool plus the ~12-second Ethereum block generation time.

Target Metric	Role	Measurement Focus	Participant 1 (HK Express: HKG → KUL)	Participant 2 (Qatar Airways: DOH → HKG)
Functional Efficiency	Data Contributor	Data-to-Asset Pipeline Execution	Completed (with friction): Required one retry due to an initial OCR formatting mismatch with the budget airline's ad-heavy, unstructured PDF layout.	Completed: Successfully uploaded, parsed, and tokenized on the first attempt, smoothly handling the complex multi-page layout of the international carrier.
Accuracy		Off-Chain Feature Engineering	With Partial Friction: Primary variables extracted correctly. However, heuristic fallback logic was required to parse an unstructured flight date format.	Successful: Required variables extracted accurately. The <code>server.py</code> pipeline successfully bypassed an edge-case multi-currency anomaly (QAR/HKD) in the tax breakdown.
System-Wide Usability	Data Consumer	API Gateway Query Execution	Successful: Insight fetched and displayed correctly on the Enterprise UI.	Successful: Insight fetched and displayed correctly on the Enterprise UI.
System Latency		End-to-End Enterprise Query Delay	~14.5 seconds (Visibly stalled waiting for ETH block generation).	~14.2 seconds (Visibly stalled waiting for ETH block generation).

Table 11 Quantitative Evaluation Results and Execution Friction in Beta Testing

3.6.2.2 Qualitative Evaluations

The unstructured think-aloud sessions revealed critical usability and psychological barriers that performance metrics alone could not capture. Feedback was synthesized based on the participants' observed behaviors, mapped directly to their domain expertise and inherent biases regarding the Web3 ecosystem.

Participant 1: The Web3 Enthusiast & HKU undergraduate

Regarding economic and societal value, Participant 1 noted that the platform effectively validates a model of economic democratization. By shifting the user from a passive data source to an active stakeholder, the system creates a direct link between digital behavior and financial reward that is missing in traditional Web2 ecosystems.

In terms of design strengths, she praised the dashboard for its legibility, noting that it successfully humanizes complex on-chain assets for everyday users without requiring a specialized background in finance or cryptography.

Participant 1 evaluated the system with inherent trust in the cryptographic backend, focusing primarily on the viability of the tokenomics and decentralized incentive structures.

- **Concern 1: Reward Tangibility vs. Token Volatility**

Participant 1 reacted highly positively to the fiat-conversion display on the Earnings Dashboard, noting it effectively gamified the data contribution process. However, she raised a critical Web3 economic concern regarding token volatility. She noted that the actual purchasing power of the tokens earned from uploading her data could arbitrarily drop overnight, a risk absent in standard Web2 reward platforms.

- **Concern 2: The Micro-Transaction Gas Flaw**

Participant 1 identified a structural economic flaw in the decentralized model. She correctly pointed out that the Ethereum mainnet gas fees required to actually claim her earned tokens to a personal wallet may easily exceed the micro-reward earned from a single flight ticket during high network congestion.

Participant 2: The Web3 Skeptic & AI Software Engineer

Participant 2 identified the correction of information asymmetry as a significant societal benefit, as the platform disrupts the monopoly of centralized data brokers. From a technical perspective, he appreciated the "clean-room abstraction" of the system architecture, which successfully hides the "architectural plumbing" of the blockchain and allows users to focus on data insights rather than smart contract execution.

Despite these strengths, his technical scrutiny led to three concerns, focusing on data privacy boundaries, system latency, and statistical transparency.

- **Concern 3: Opacity in Data Extraction**

Approaching the system with high skepticism, Participant 2 exhibited visible hesitation before confirming the upload of his Qatar Airways ticket. He questioned how he could be certain the Python server did not also scrape sensitive information like passport numbers or home addresses from the PDF. This highlighted that skeptical users require a transparent intermediate UI state that explicitly highlights exactly which text strings were extracted before cryptographic consent is requested.

- **Concern 4: The Demand for Algorithmic Explainability**

Leveraging his statistics and AI background, Participant 2 heavily criticized the black box nature of the on-chain ML output when acting as the enterprise consumer. He noted that simply receiving an immutable, mathematically bounded risk score is useless for a regulated financial institution, which must be able to statistically explain why a user was penalized or approved to comply with strict lending regulations.

- **Concern 5: UI Intuitiveness vs. Blockchain Latency**

As a software engineer, the user praised the API Gateway's Web2-style interface, noting it elegantly abstracted the Solidity backend. However, he explicitly stated that the observed 14-second wait time for the blockchain query to resolve renders the system entirely incompatible with automated, high-frequency enterprise underwriting systems that expect millisecond API responses.

3.6.2.3 Internal Real-World Data Demonstration

Prior to external Beta Testing, an internal end-to-end trial was conducted to validate the data ingestion and feature engineering pipelines. A real Cathay Pacific e-ticket receipt was utilized as the baseline document for this evaluation.

Raw text was successfully extracted by the `server.py` backend pipeline utilizing OCR, which was then seamlessly mapped to the system's engineered features. Through applied heuristic rules, the flight cost and booking class were mathematically downscaled into the `creditAmount` proxy, while the flight frequency data was mapped to the `tripsAnnual` integer. Through this internal demonstration, it was successfully verified that complex off-chain documents could be accurately translated into the strict `uint256` numerical constraints required by the on-chain Decision Tree Oracles, establishing the baseline stability required to proceed with external Beta trials.

Overall, the synthesis of quantitative and qualitative findings confirms that DataTrek is fully capable of transforming real-world behavioral data into secure, monetizable assets. However, the evaluation also proves that raw technical functionality is insufficient for widespread adoption. While pragmatic users are hindered by blockchain latency and gas fee economics, privacy-conscious users remain hesitant due to Web3 wallet friction and a lack of algorithmic transparency.

In summary, this evaluation validates that the platform successfully achieves (1) high predictive accuracy and (2) extreme computational efficiency within the constraints of the EVM. Conversely, the transition to real-world data exposed critical bottlenecks in (3) network latency and (4) gas-to-reward economics, while qualitative feedback highlighted a demand for (5) algorithmic transparency. Collectively, these findings validate the core decentralized value proposition while providing a clear engineering roadmap for addressing the infrastructural hurdles of the current Web3 landscape.

3.7 End-to-End Application Walkthrough (User Manual)

Section 3.7 provides a step-by-step user manual for the platform, detailing the workflows for two primary user roles: **Data Contributors** (individuals monetizing their personal data) and **Data Consumers** (enterprises purchasing verified behavioral insights).

The walkthrough is structured as follows: Section 3.7.1 (Universal Onboarding) details the unified Web2-to-Web3 login and MetaMask wallet authentication for all users. Section 3.7.2 (Data Contributor Guide) explains how individuals upload data via AI extraction, govern buyer access requests, and redeem platform rewards for ETH. Section 3.7.3 (Data Consumer Guide) outlines how enterprises navigate the marketplace, purchase access via smart contract escrows, and unlock DPIs. Section 3.7.4 (External Enterprise Access): demonstrates how off-platform partners use the Secure API Portal to query metadata and execute on-chain machine-learning algorithms.

3.7.1 Universal Onboarding (Both All Users)

Before dividing into specific roles, all users share a simple, unified login process to bridge the gap between traditional Web2 accounts and Web3 blockchain security.

1. **Social Login:** Navigate to the platform homepage. To avoid complex blockchain setups right away, sign in using a familiar Social SSO provider (e.g., Gmail, Microsoft, or Facebook) (see Figure 23).

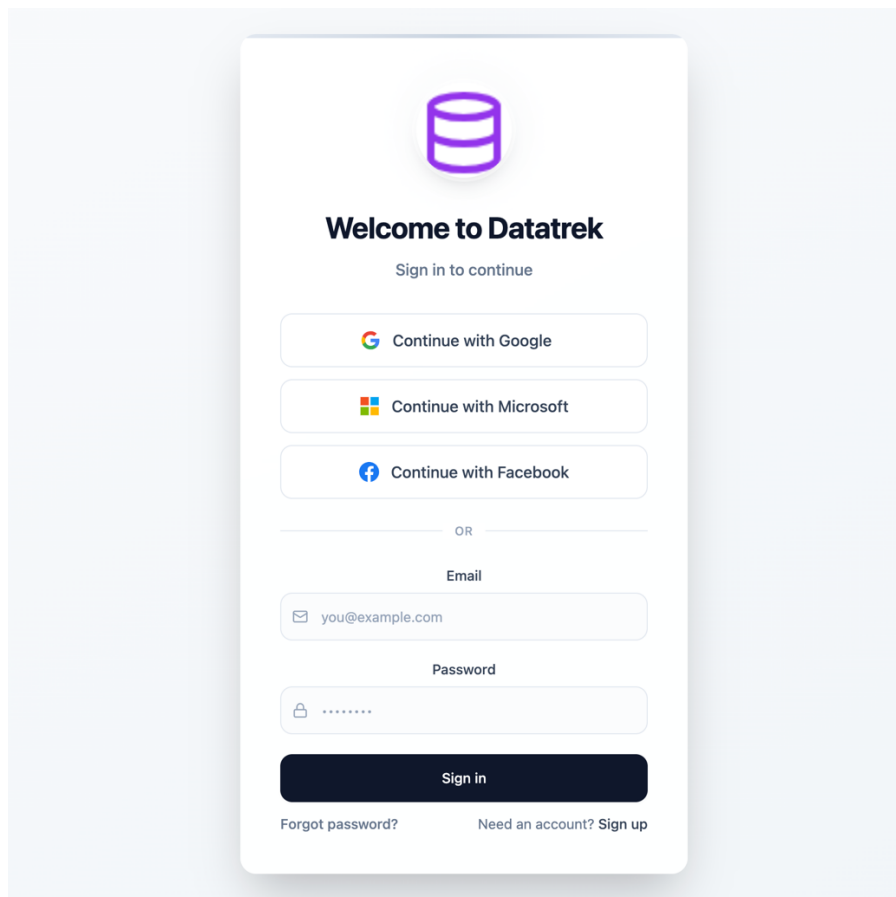


Figure 23 Unified Social SSO Onboarding Interface

2. **Connect Wallet:** Once logged in, look to the top-right corner of the navigation bar and click "**Connect Wallet**" (see Figure 24(a)).
3. **Authorize MetaMask:** A MetaMask pop-up will appear. Ensure your wallet is set to the **Sepolia Testnet** and approve the connection (see Figure 24(b)). Your session is now securely authenticated on the blockchain, and the button will update to display your shortened wallet address (see Figure 24(c)).

Figure 24 Web3 Wallet Onboarding and MetaMask Authorization Sequence

(a)

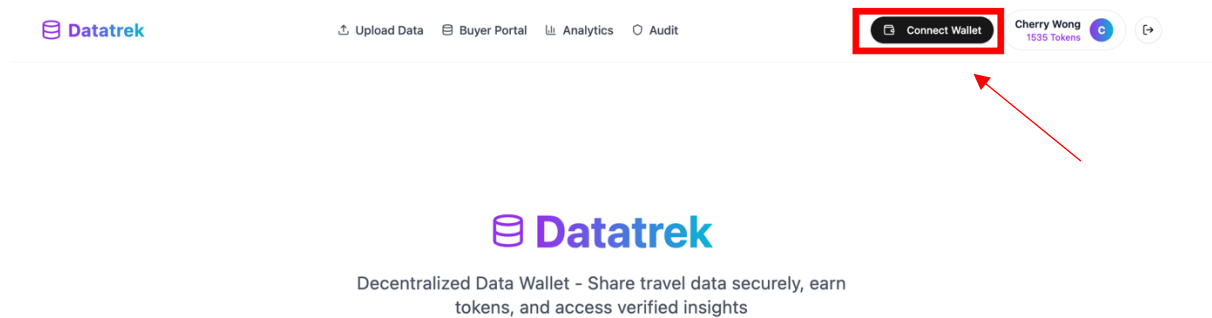


Figure 24(a) Initiation of the Web3 connection via the "Connect Wallet" call-to-action.

(b)

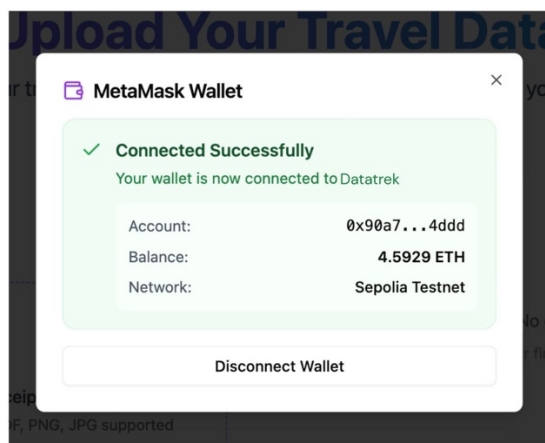


Figure 24(b) Successful authorization and connection confirmation within the MetaMask wallet

(c)

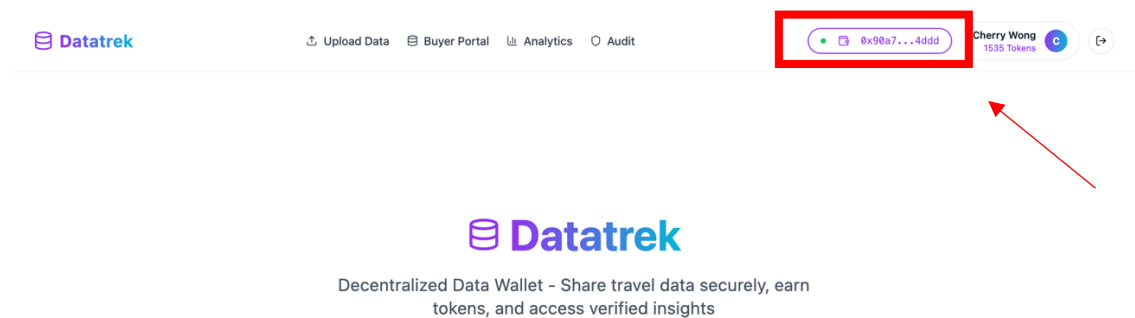


Figure 24(c) Resulting authenticated status on DataTrek, displaying the contributor's shortened public wallet address

3.7.2 Data Contributor Guide: Ingestion, Governance, and Rewards

As a Data Contributor, your goal is to upload your behavioral data, control who gets to see it, and earn rewards when enterprises purchase your insights.

Step 1: Uploading Your Data

1. On your main dashboard, click the **"Start Uploading"** button (see Figure 25).
2. Drag and drop your raw data file (eg. PDF/ PNG/ JPEG) into the upload zone.
3. Click **"Upload and Analyse"**.

The screenshot shows the Datatrek dashboard with a navigation bar at the top containing 'Upload Data', 'Buyer Portal', 'Analytics', and 'Audit'. The user 'Cherry Wong' has 1535 Tokens. The main heading is 'Upload Your Travel Data' with the subtext 'Share your travel experiences securely and earn tokens. Control your privacy and contribute to better insights.' A red arrow points to a red-bordered box containing the 'Upload Travel Data' section. This section includes a drop zone for tickets or receipts, a file upload area showing 'CHOW KA HO PROF 30SEP HKG.pdf' (0.73 MB), and a button 'Reading your document with AI...' (Extracting flight details, dates, and payment info). Below these are form fields for 'Full Name *', 'Email Address *', 'Trip Name *', 'Category', and a 'Description' text area. To the right, the 'Data Preview' section shows 'No data uploaded yet'. Below that, 'Data Quality & Market Value' displays a 'Completeness Score' of 0% and a 'Market Demand' of 'Low'. The 'Suggested Actions' section lists two items: 'Add more travel segments to improve completeness' and 'Upload diverse data types for higher rewards'. At the bottom, the 'Privacy & Permissions' section has a toggle for 'Allow Anonymization'. The 'Token Rewards' section shows 'Total Earned' as 0 Tokens and 'Active Assets' as 0 Datasets.

Figure 25 AI-Assisted Data Ingestion and Metadata Entry Interface

***Note:** The system's AI will automatically perform OCR to extract the relevant info from your document*

Step 2: Data Valuation Once uploaded, the platform’s backend instantly analyzes your file without exposing the raw contents (see Figure 26). The screen will update to show:

- **Document Verification:** The system automatically verifies the authenticity of the file and data uploaded.
- **Completeness Score:** A percentage showing how readable and complete your data is.
- **Market Demand:** An indicator showing how highly sought-after your specific data type is by current buyers.

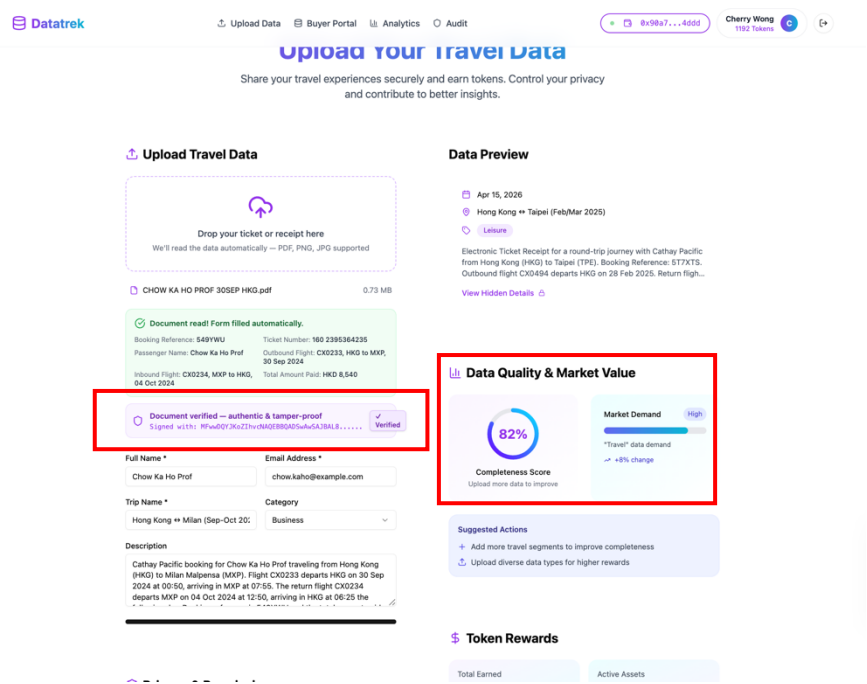


Figure 26 Successful Data Verification, Quality Assessment, and Market Valuation

Note: The interface displays the post-upload state, showcasing successful AI data extraction, cryptographic document verification, and the calculated completeness score.

Step 3: Approving Buyer Requests (Bidirectional Consent)

You have total control over your digital assets. When an enterprise wants to purchase access to your DPIs — specifically your calculated Alternative Credit Score and Health Index—the platform does not grant access automatically. Crucially, buyers are only purchasing access to these mathematical insights, *never* your raw personal data.

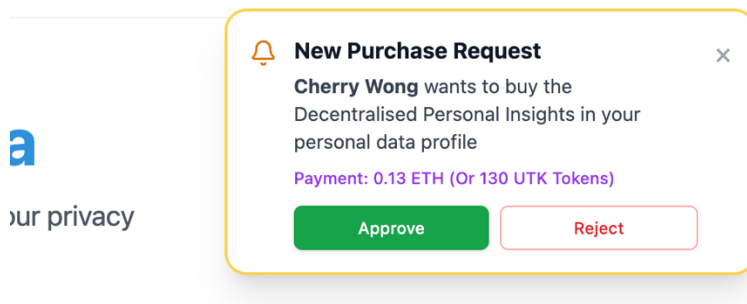


Figure 27 Data Contributor-Side Data Access Request Notification

1. You will receive a "**New Purchase Request**" pop-up on your dashboard detailing which enterprise wants to access your DPIs and their specific business reason for doing so (see Figure 27).
2. You can review the details of the request and choose to either **Approve** or **Reject** it.
3. If you click **Approve**, the smart contract securely unlocks your Credit Score and Health Index for the buyer to view, and instantly deposits the payment into your account.

Step 4: Redeeming Rewards for ETH

Once you have accumulated platform rewards from data sales, you can claim them as liquid cryptocurrency (Sepolia ETH) directly into your personal wallet.

1. Navigate to the Token Rewards Module and click **"Redeem to Sepolia,"** which diverts you to the dedicated Sepolia Faucet page for redeeming your earned tokens to SepETH.

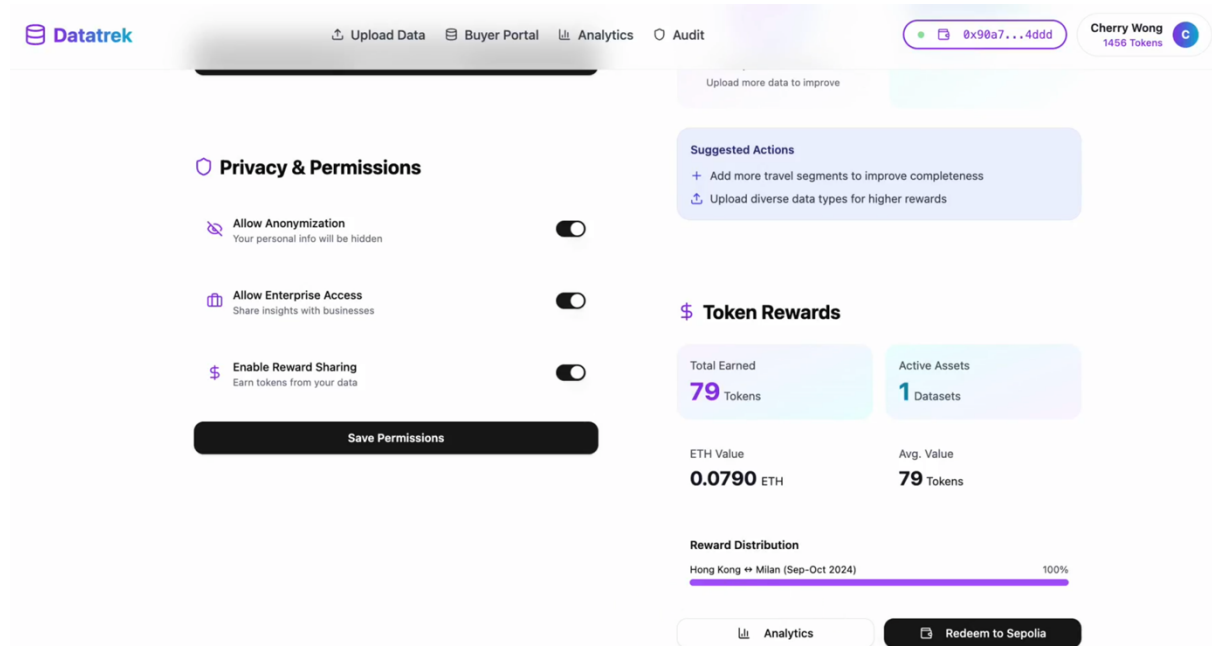


Figure 28 Reward Liquidation Interface – Redeem to Sepolia ETH

2. If your wallet is not already linked on this page, click the blue **"Connect Wallet"** button to establish the secure connection (see Figure 29 (a)).
3. Enter the specific amount of testnet ETH you wish to redeem in the input field (up to the allowed maximum).
4. Click the green **"Claim Sepolia ETH"** button (see Figure 29 (b)).
5. **MetaMask Confirmation:** A MetaMask pop-up will appear asking you to confirm the network transaction. *(Note: As indicated directly on the interface, MetaMask will say "Sending 0 ETH" — this is completely normal, as you are simply authorizing the smart contract to release funds to you).*
6. The interface will temporarily display a **"Waiting for blockchain confirmation..."** status while the network processes your request (see Figure 29 (c)).

7. Once finalized, a green **Success Notification** will appear at the bottom of the module, and the SepETH will instantly be deposited into your MetaMask wallet! (see Figure 29 (d))

Figure 29 Reward Liquidation Interface – Claiming Sepolia ETH via Faucet

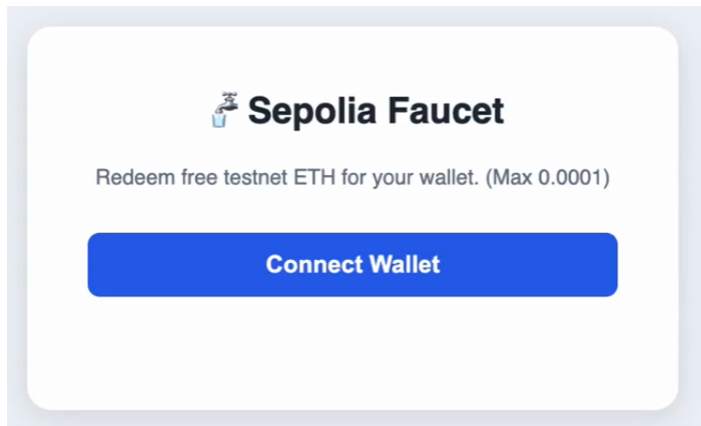


Figure 29(a) Initial Faucet Interface and Wallet Authorization

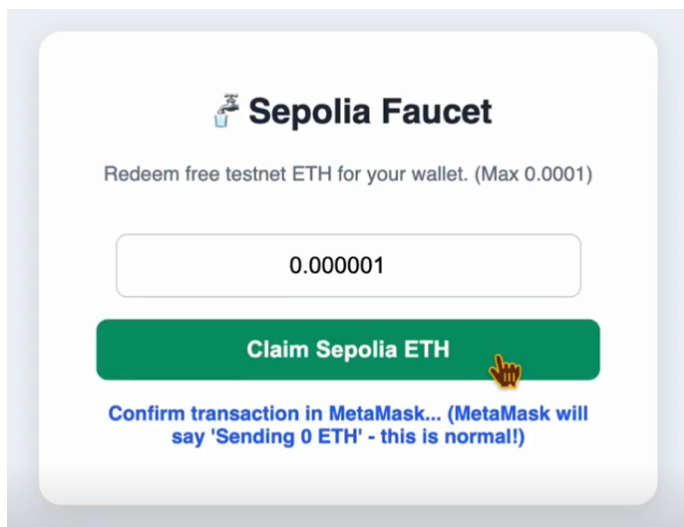


Figure 29(b) Reward Amount Specification and Smart Contract Prompt

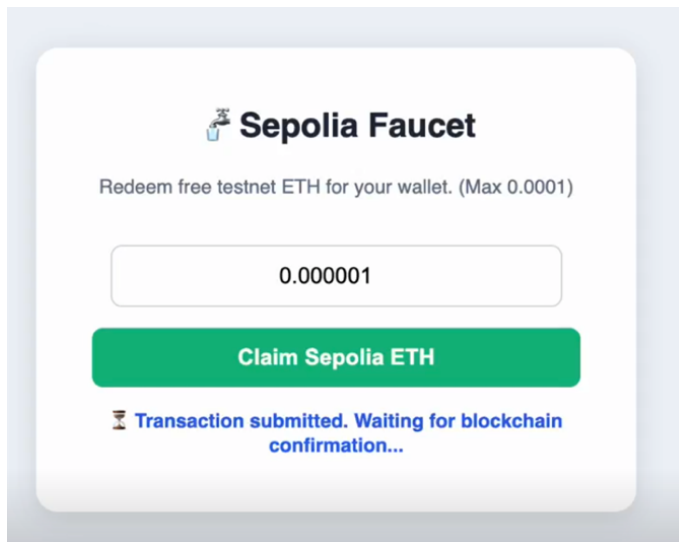


Figure 29(c) Asynchronous Transaction Pending State

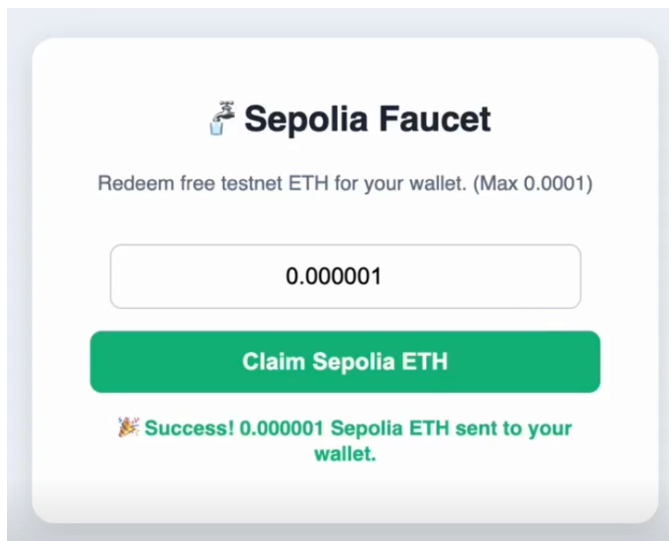


Figure 29(d) Successful Liquidation and On-Chain Verification

3.7.3 Data Consumer Guide: Discovery and Acquisition

As a Data Consumer (Enterprise), your primary objective is to browse the decentralized marketplace to find verified, predictive machine-learning insights—such as Alternative Credit Scores and Health Indexes. Crucially, this platform utilizes a "Privacy-by-Design" architecture. You are purchasing access to DPIs, not the user's raw, underlying data.

Step 1: Browsing Individual's Data Profile in Marketplace

1. After completing the Universal Onboarding, ensure you are in the **"Data Consumer"** role. Select the **"Buyer Portal"** tab from the main header.
2. You will arrive at the **Data Buyer Portal**, which acts as your central interface for discovery (see Figure 30).

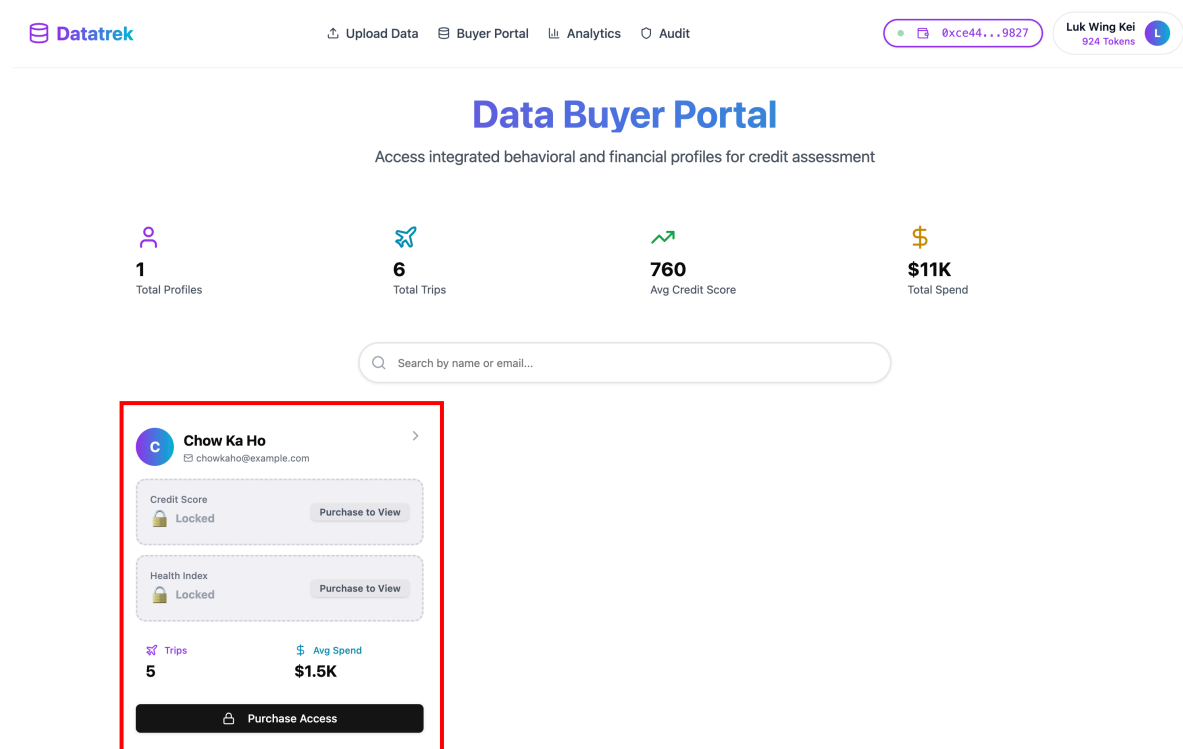


Figure 30 Data Buyer Portal and Data Profile Marketplace Discovery

3. The portal immediately provides **Aggregated Network Statistics** at the top of the screen. This allows you to quickly gauge the commercial potential of the available data pool.
4. Below these statistics is a grid of anonymous **User Data Profile** (e.g., the profile for "Chow Ka Ho").

5. To uphold user privacy, the valuable predictive insights — the **Alternative Credit Score** and **Health Index** — feature a "Locked" padlock icon by default.

Step 2: Initiating a Secure Purchase

1. Once you identify a promising profile, click the **"Purchase Access"** button on that specific User Card (see Figure 31).
2. A **Confirm Data Access Transaction** window will appear.
3. **Smart Contract Audit:** For total transparency, you can click "View on SepoliaScan" to audit the exact blockchain contract handling your transaction.
4. **Business Use Case:** You must type a professional justification into the "Business Use Case" field (e.g., "view credit score"). This input is recorded on the blockchain, creating a permanent, transparent audit trail of why the data was accessed.
5. **Cost Breakdown:** Review the financial parameters at the bottom. The system clearly displays the Total Cost (e.g., 0.13 ETH), the Platform Fee (5%), and the net amount the individual data contributor will receive.
6. Click **"Sign & Commit Transaction."** MetaMask will prompt you to authorize the payment.

Confirm Data Access Transaction

Chow Ka Ho
chowkaho@example.com
6 Travel Profiles

Credit Score: Purchase Access to View
Total Trips: 6 (Public Info)

Smart Contract Address
0x71C7656EC7ab88b098defB751B7401B5f6d8976F
[View on SepoliaScan](#)

Business Use Case *
Explain how you'll use this travel data for enterprise analysis, credit scoring, or business insights...

Total Cost:	0.13 ETH (or 130 UTK Tokens)
Platform Fee:	5%
Seller Receives:	0.1235 ETH

Sign & Commit Transaction

Escrow Protection: Your funds will be locked in the smart contract until the seller approves. If rejected, funds are automatically returned.

Figure 31 Navigate to the Token DPI Access Confirmation and Financial Breakdown in Purchase Initiation

Step 3: Checkout and Escrow Protection

1. Once you authorize the payment in MetaMask, the transaction moves into a secure holding state.
2. The interface will update to display **"Waiting for Seller Approval..."**(see Figure 32).
3. Because the platform enforces bidirectional consent, your funds are safely locked in a smart contract escrow. The DPI is *not* released, and the funds are *not* transferred, until the individual data contributor manually approves your request on their dashboard. If they reject it, your funds are automatically returned.

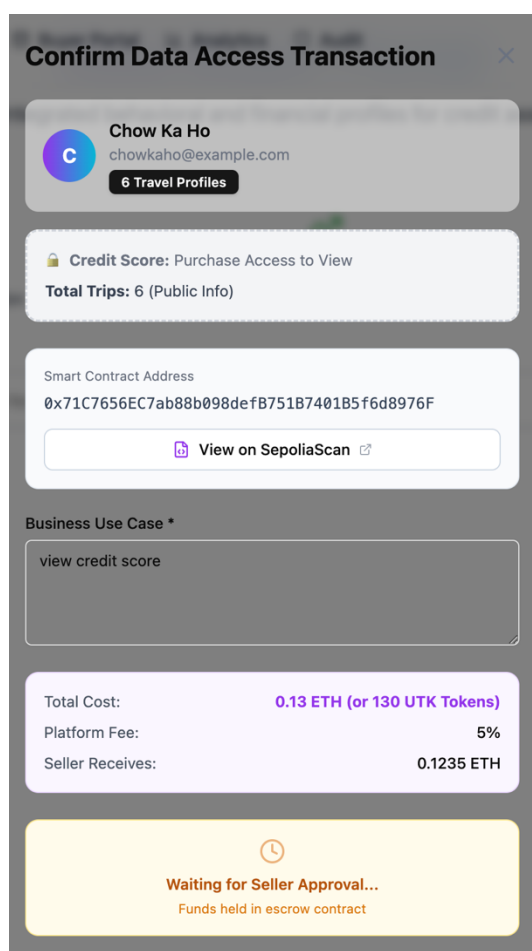


Figure 32 Smart Contract Escrow and DPI Pending Approval State

Step 4: Unlocking the Insights

1. The moment the individual data contributor clicks “Approve,” the smart contract executes the exchange instantly.
2. Your screen will update with a green “Access Granted! Data unlocked successfully” notification.
3. The modal will close, and your dashboard will automatically refresh.
4. On the User Card, the “Locked” padlocks are now gone. The mathematical insights are fully decrypted and visible (e.g., revealing a Credit Score of 355 and a Health Index of 95/100) (see Figure 33).

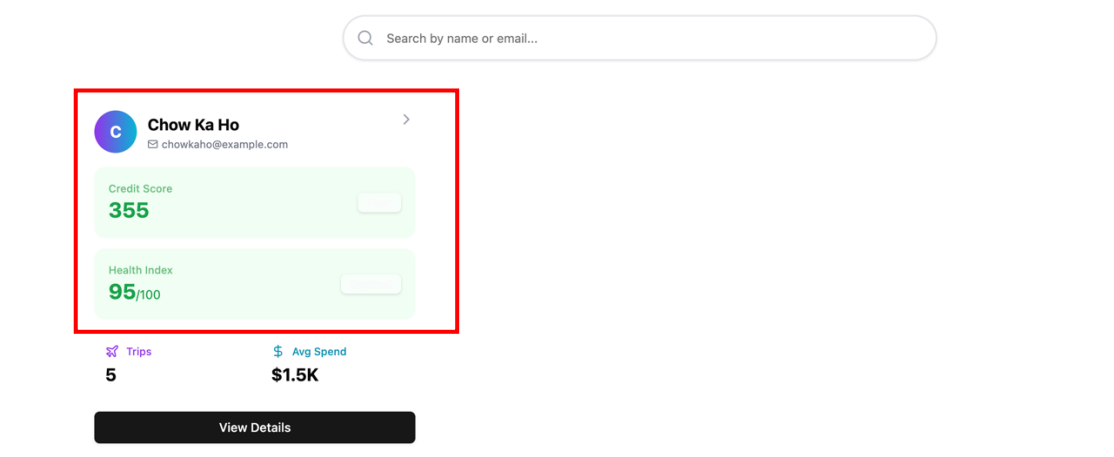


Figure 33 Refreshed Data Consumer Interface with Unlocked Predictive DPI of an Individual’s Data Profile

5. The button on the card changes from “Purchase Access” to “**View Details.**”

Step 5: Analyzing the Comprehensive DPI and Relevant DPI Report

1. Click "**View Details**" to open the final output of the transaction.
2. You now have access to a highly granular, enterprise-grade **Credit Assessment and Behavioral Profiling Report** (see Figure 34).
3. This dashboard breaks down the user's travel metadata into strategic analytical categories to aid in enterprises' business decision-making.

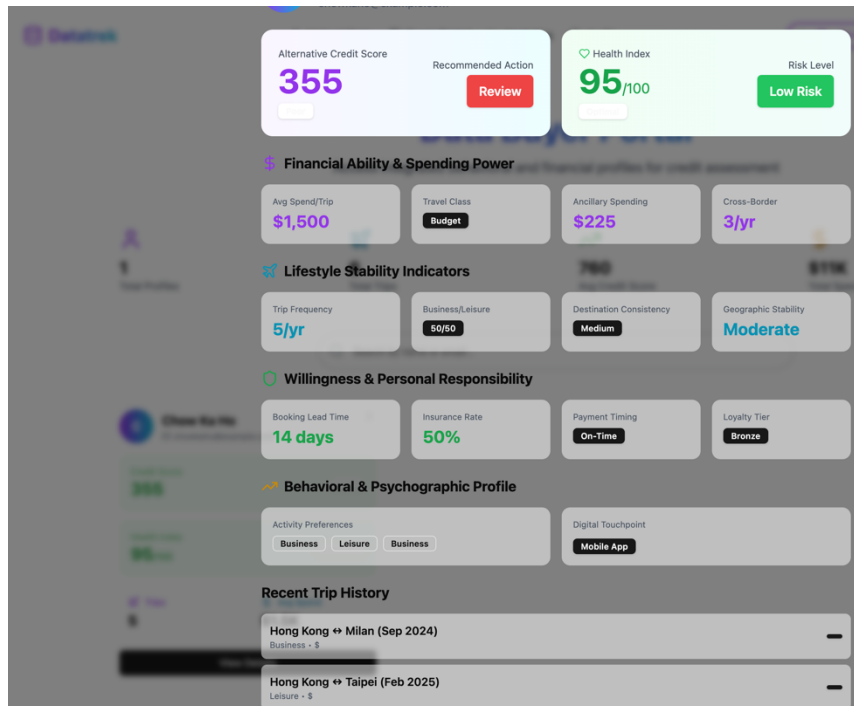


Figure 34 Unlocked Multi-Dimensional Behavioral Profiling and On-Chain Credit Evaluation Report

3.7.4 External Enterprise Access via Secure API Portal

DataTrek also supports external enterprises and institutional partners who may not be direct, active users of the platform's primary marketplace but still require secure access to verified insights. If an enterprise has purchased off-platform access, they can securely retrieve DPIs using our dedicated API gateway.

Step 1: Secure API Authentication

1. Navigate to the **Private Data Portal**, which serves as the dedicated enterprise gateway.
2. Enter the target **Data Owner Name** (e.g., "Chow Ka Ho") whose profile you have been authorized to review (see Figure 35).
3. Enter your enterprise **API Key** into the secure field (e.g., entering the secure phrase "base44").
4. Click **"Unlock Data."** The system will securely connect via the API and decrypt the user's base, non-identifying behavioral metadata (such as Mapped Age, Mapped Credit Spend, and Annual Trips) onto the left side of your resulting dashboard.

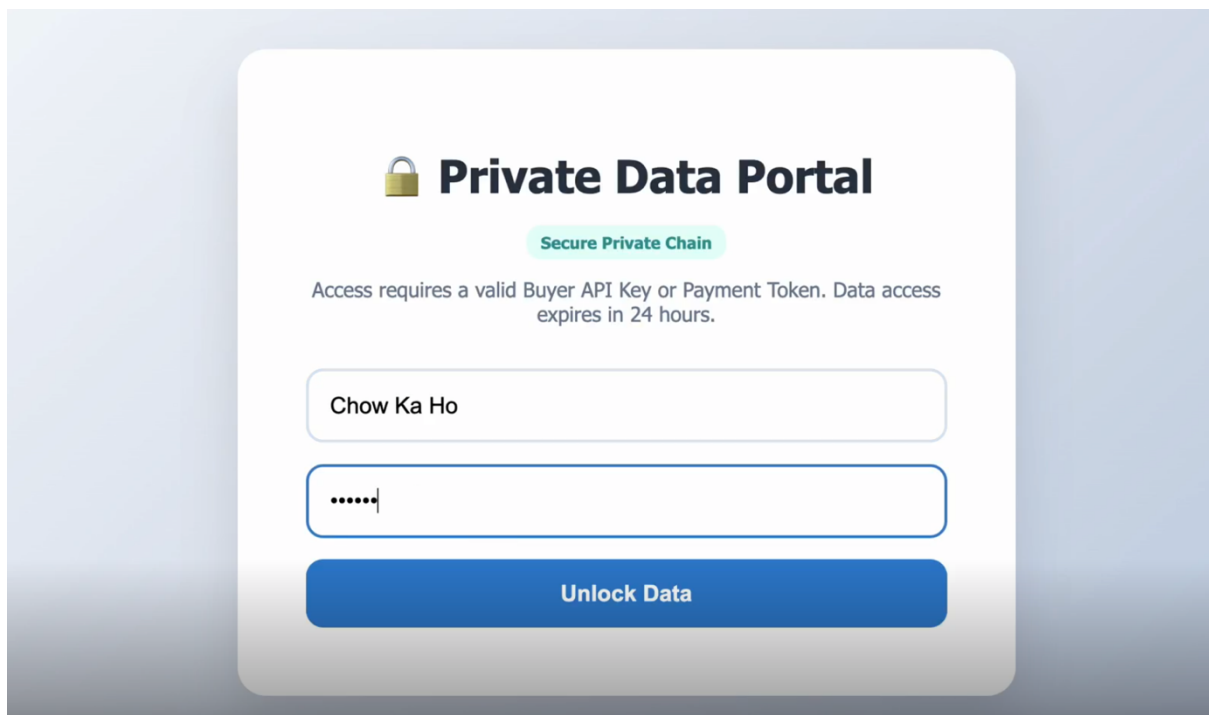
The image shows a web interface for the 'Private Data Portal'. At the top, there is a padlock icon followed by the title 'Private Data Portal'. Below the title is a green pill-shaped button labeled 'Secure Private Chain'. A line of text states: 'Access requires a valid Buyer API Key or Payment Token. Data access expires in 24 hours.' There are two input fields: the first contains the text 'Chow Ka Ho', and the second contains a series of dots for a password. Below these fields is a large blue button labeled 'Unlock Data'.

Figure 35 Enterprise API Gateway and Secure Authentication Portal

Step 2: On-Chain ML Algorithms Model Execution

Immediately after entering the correct Data Owner Name and API Key and clicking "Unlock Data," you are redirected to the Pre-Execution Interface (see Figure 36). Even though you are accessing the data via an external API, DataTrek ensures that the actual machine-learning execution remains decentralized, deterministic, and mathematically verifiable on the blockchain.

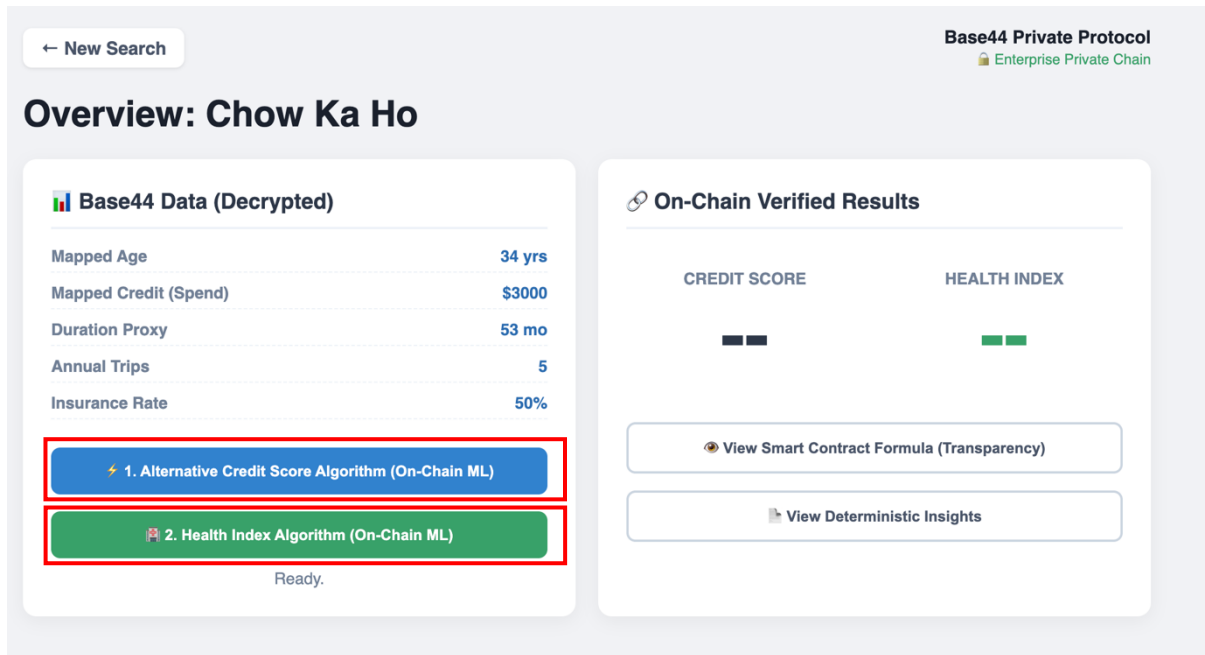


Figure 36 Pre-Execution Dashboard Displaying Encrypted DPIs

1. On the decrypted interface, click the **"1. Alternative Credit Score Algorithm (On-Chain ML)"** button.
2. **Transaction Confirmation:** A MetaMask pop-up will appear. You must confirm the transaction and pay the required gas fee to execute the smart contract directly on the Sepolia testnet (see Figure 37).

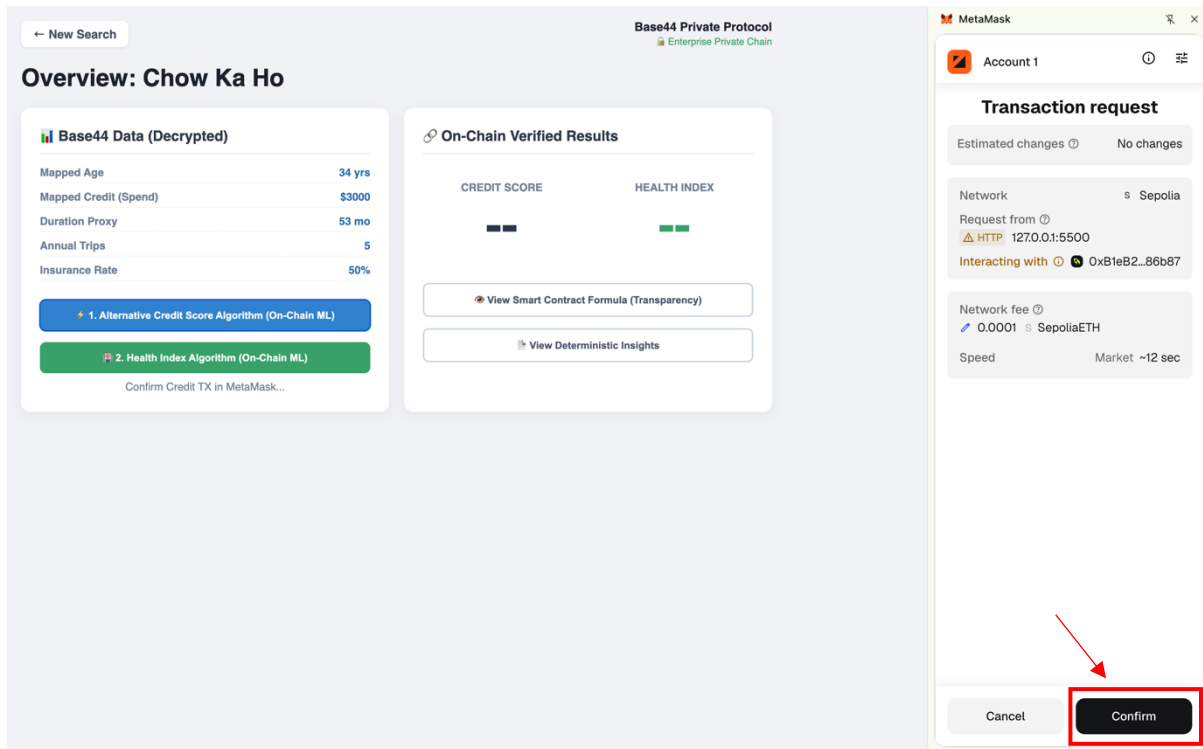


Figure 37 On-Chain Model Execution and MetaMask Smart Contract Authorization

3. Once the transaction is mined on the blockchain, the system will execute the computation and the exact score (e.g., 355 / Poor) will dynamically appear on the right panel under "On-Chain Verified Results."
4. Next, click the **"2. Health Index Algorithm (On-Chain ML)"** button and confirm the secondary MetaMask transaction.
5. Upon successful execution, the interface will display a green "Health Insight Updated!" notification, and the user's computed Health Index (e.g., 95/100 / Optimal) will be revealed (see Figure 38).

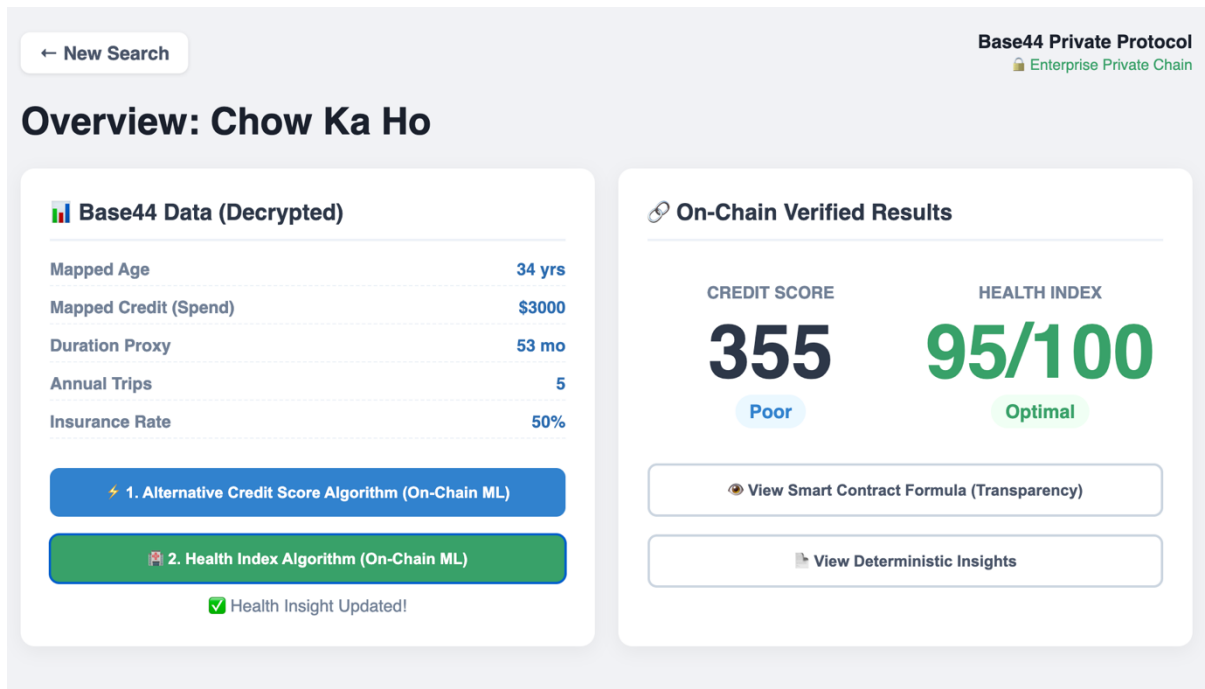


Figure 38 Executed State Displaying Verified On-Chain Alternative Credit Score and Health Index

Step 3: Algorithmic Transparency Check

To uphold DataTrek's core "Whitebox AI" philosophy, external consumers can audit the exact mathematical logic of the machine learning algorithms at any time.

1. Click the **"View Smart Contract Formula (Transparency)"** drop-down button located beneath the generated scores.
2. The interface will instantly display the raw, on-chain conditional logic (the exact EVM code) used to calculate both the Alternative Credit Score Algorithm and the Health Index Algorithm based on the user's metadata. This provides absolute proof that the scores were generated fairly, deterministically, and without centralized tampering.

Step 4: Accessing Deterministic Insights

1. To compile the decrypted metadata and the newly generated on-chain scores into a standardized, business-ready format, click the **"View Deterministic Insights"** button.
2. The system will process the inputs and generate a comprehensive **Data Insights Analysis** report tailored specifically to that individual user (see Figure 39).

3. This streamlined report categorizes the user's profile into actionable enterprise underwriting metrics, clearly displaying indicators for *Financial Ability*, *Lifestyle Stability*, and *Willingness & Risk*, concluding with a final automated recommendation.
4. Click "**Back to Home**" at the bottom of the report to securely clear the session and return to the main API portal.

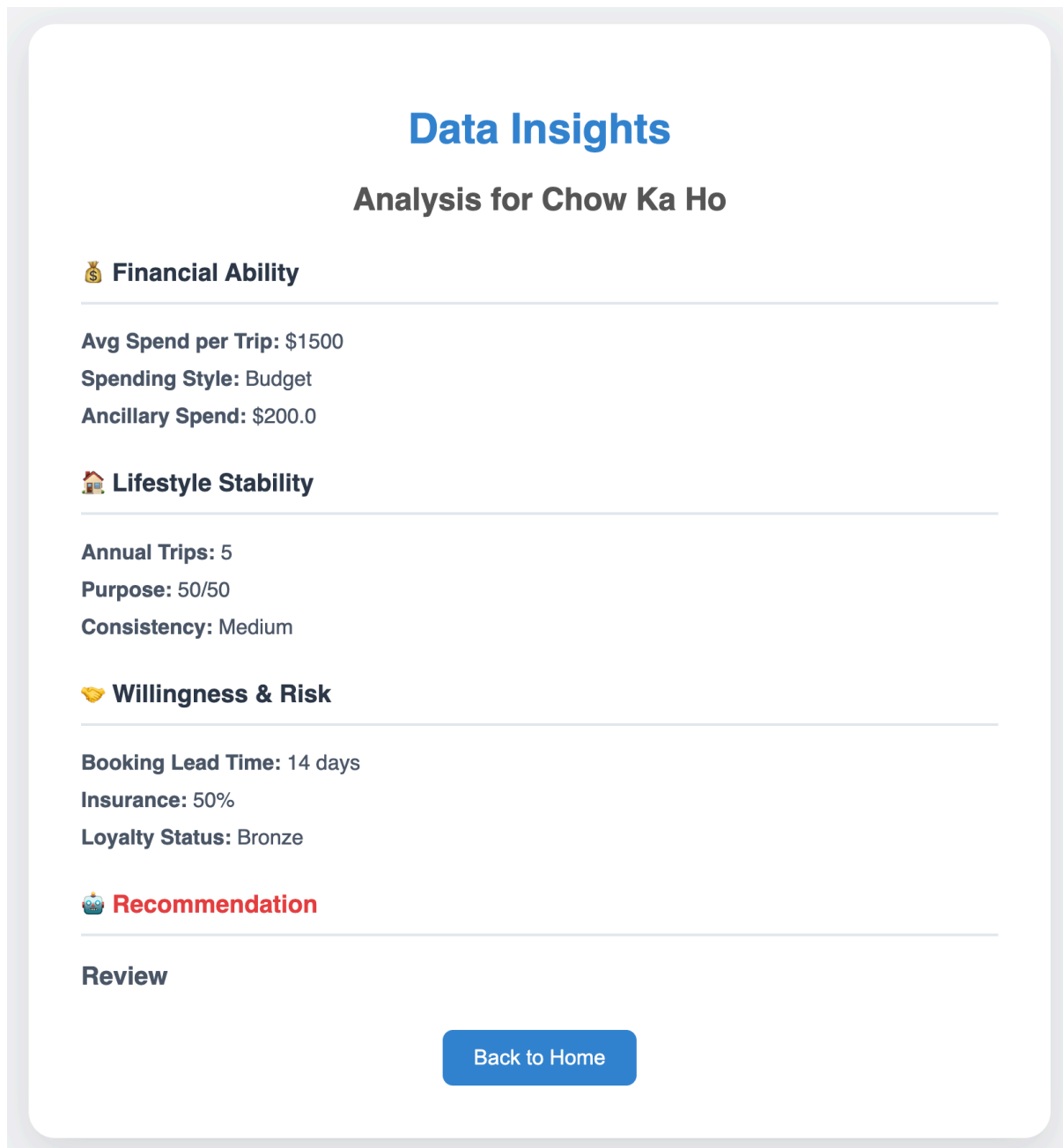


Figure 39 Additional Detailed Data Insights Analysis and Enterprise Risk Report generated further from DPI

4. Conclusion, Limitations, and Future Work

4.1 Conclusion

The DataTrek prototype successfully demonstrates the technical feasibility of a decentralized, privacy-preserving "Contribute-to-Earn" ecosystem by transforming complex behavioral data into verifiable digital assets without compromising user sovereignty. By transitioning from an initial hybrid design to a finalized native on-chain ML architecture, the project establishes a robust, trustless environment where users are fairly and transparently incentivized for their data contributions. The integration of off-chain preprocessing pipelines and IPFS storage with the EVM for deterministic ML execution provides a scalable blueprint that bridges the gap between traditional centralized data brokering and modern decentralized finance.

4.2 System Limitations

Despite its robust foundational architecture, the prototype faces three critical limitations inherent to current Web3 deployments. First and foremost is the legal blocker regarding regulatory compliance; the permanent immutability of on-chain data, such as the STB tokens, fundamentally clashes with the "Right to be Forgotten" mandated by privacy frameworks like General Data Protection Regulation (GDPR), restricting widespread enterprise adoption. Secondly, the system is constrained by economic scalability and execution latency. Executing native ML inferences on the Ethereum mainnet incurs gas fees that can spike unpredictably, while the ~12-second block generation time introduces a synchronous delay that renders the platform unviable for automated, high-frequency underwriting APIs. Finally, the architecture currently relies on a centralized Python-based OCR scanner to read PDFs before sending metadata on-chain. This creates an inherent "Oracle Problem" - if this off-chain ingestion point is compromised or makes a parsing error, the smart contract will blindly execute false data, thereby jeopardizing the trustless guarantee of the entire decentralized pipeline.

4.3 Future Work

To overcome current architectural bottlenecks and prepare the platform for commercial deployment, future iterations will focus on four orthogonal research vectors: decentralized governance, network scalability, advanced cryptographic privacy, and the decentralized crowdsourcing of machine learning algorithms. Firstly, the hardcoded smart contracts will be transitioned into a Decentralized Autonomous Organization (DAO) featuring a community voting system. This will empower token holders to actively propose and vote on protocol upgrades, reward ratios, and new data parameters. Secondly, to ensure economic viability and near-instant API latency, the protocol will be optimized for Ethereum L2 solutions to drastically reduce the gas costs of on-chain ML execution. Furthermore, the integration of Zero-Knowledge Proofs (ZKPs) will allow users to cryptographically prove the validity of their data insights to enterprise consumers without exposing any raw metadata to the public ledger. This advancement will effectively reconcile blockchain immutability with strict global privacy regulations.

In addition to these infrastructural upgrades, DataTrek will evolve its core analytical engine. Inspired by collective intelligence and collaborative machine learning paradigms pioneered by researchers at the MIT Media Lab [29], the platform will implement a decentralized crowdsourcing framework for its algorithmic backend. Rather than relying on a static, pre-trained model, DataTrek will open-source its ML architecture, allowing global data scientists and developers to collaboratively train, polish, and submit improved algorithms. Utilizing token-incentivized smart contract marketplaces [30], [31], developers will be able to propose more accurate or gas-optimized models. If a submitted model successfully passes automated backtesting and community consensus, it can be seamlessly plugged in to replace the legacy algorithm, earning the contributors a percentage of future marketplace royalties. Crucially, these four development pathways are mathematically and architecturally orthogonal to each other. Because they operate at distinct layers of the technology stack—consensus, execution, privacy, and application — they can be researched and implemented in parallel by different development teams without creating technical debt or cross-system dependencies.

5. Project Schedule

Time Period	Key Tasks & Technical Pivots	Status
September	Meet with supervisor to define project scope	Done
	Phase 1 Deliverables: <i>Detailed project plan</i> <i>Project website</i>	Done
	- Conduct MLR on P2P data tokenomics, blockchain architecture, and privacy-preserving techniques. - Identify suitable behavioral data domains (FinTech/InsurTech) and target insights.	Done
October	- Case Studies Review: - Decentralized P2P marketplaces - Self-sovereign identity - Decentralized AI training - Refine research questions based on MLR findings. - Develop technical protocol: - Define data types, token standards, access control logic - Build four-layered system architecture.	Done
November	PoC Development (Phase 1): frontend interface	Done
December	PoC Development (Phase 2): - Backend logic - Blockchain integration - Smart contract development for utility tokens and access control. <i>Initial on-chain ML Research:</i> - Off-chain model training utilizing Random Forest classifiers.	Done
January	PoC Development (Phase 3): - Full stack integration (connect Python backend with React frontend).	Done
	Phase 2 Deliverable: <i>Interim Report</i>	Done
February	Architectural Pivot: - Transition from off-chain aggregation to Native On-Chain ML. Engineering: - Develop Python feature engineering pipeline, transpile ML models to Solidity Decision Trees, and apply integer scaling for gas optimization.	Done

	• System refinement based on identified beta UI friction and EVM constraints.	
March	• Evaluation (Phase 1) – Alpha Testing: ○ Analyze quantitative metrics (system latency, gas cost consumption) • Evaluation (Phase 2) – Beta Testing: ○ Conduct unstructured think-aloud sessions and technical evaluations. ○ Synthesize qualitative think-aloud feedback (e.g., algorithmic opacity, micro-transaction flaws). • Finalize the DPI Generation Engine ○ Credit Oracle validation ○ Health Oracle validation	Done
April	• Write final report. • Prepare for the final presentation. • Finalize project website.	Done
	• Phase 3 Deliverable ○ <i>Final report</i> ○ <i>Final presentation</i>	Done

References

- [1] K. Birch, D. T. Cochrane, and C. Ward, "Data as asset? The measurement, governance, and valuation of digital personal data by Big Tech," *Big Data & Society*, vol. 8, no. 2, pp. 1–15, Jul.–Dec. 2021. [Online]. Available: <https://doi.org/10.1177/20539517211017308>.
- [2] World Economic Forum, "Personal Data: The Emergence of a New Asset Class," *World Economic Forum ITTC Report*, Jan. 2011. [Online]. Available: https://www3.weforum.org/docs/WEF_ITTC_PersonalDataNewAsset_Report_2011.pdf. [Accessed Sep. 21, 2025].
- [3] S. Zuboff, *The Age of Surveillance Capitalism: The Fight for a Human Future at the New Frontier of Power*. New York, NY: PublicAffairs, 2019.
- [4] Pew Research Center, "Americans and Privacy: Concerned, Confused and Feeling Lack of Control Over Their Personal Information," Washington, DC, USA, Nov. 15, 2019. [Online]. Available: <https://www.pewresearch.org/internet/2019/11/15/americans-and-privacy-concerned-confused-and-feeling-lack-of-control-over-their-personal-information/>. [Accessed Sep. 22, 2025].
- [5] T. Christl, "How companies use personal data against people – Working Paper," Cracked Labs, Vienna, Austria, Oct. 2017. [Online]. Available: https://crackedlabs.org/dl/CrackedLabs_Christl_DataAgainstPeople.pdf [Accessed Nov. 30, 2025].
- [6] M. M. Merlec and H. P. In, "DataMesh+: A blockchain-powered peer-to-peer data exchange model for self-sovereign data marketplaces," *Sensors*, vol. 24, no. 6, p. 1896, Mar. 2024. [Online]. Available: <https://doi.org/10.3390/s24061896>.
- [7] Ocean Protocol Foundation, "Meet Ocean: Tokenized AI & Data — Ocean Protocol." Accessed Jan. 24, 2026. [Online]. Available: <https://oceanprotocol.com/>
- [8] V. Baraku, E. Ramadani, I. Paraskakis, S. Veloudis, and P. Yadav, "Defining personal data sovereignty: An ontologically-based framework facilitating subject privacy control," *Data and Information Management*, vol. 9, no. 1, p. 100108, 2025. [Online]. Available: <https://doi.org/10.1016/j.dim.2025.100108>.
- [9] M. S. Mayernik et al., "Data Conservancy Provenance, Context, and Lineage Services: Key Components for Data Preservation and Curation," *Data Science Journal*, vol. 12, pp. 158–171, 2013. [Online]. Available: <http://doi.org/10.2481/dsj.12-039>.

- [10] C. P. Chen and C. Y. Zhang, "Data-intensive applications, challenges, techniques and technologies: A survey on Big Data," *Information Sciences*, vol. 275, pp. 314–347, Aug. 2014. [Online]. Available: <https://doi.org/10.1016/j.ins.2014.01.015>
- [11] World Bank Group, "The use of alternative data in credit risk assessment: Opportunities, risks, and challenges," *World Bank*, Washington, DC, USA, 2024. [Online]. Available: <http://documents.worldbank.org/curated/en/099031325132018527>. [Accessed: Sep. 25, 2025].
- [12] CGAP, "Leveraging Transactional Data for Micro and Small Enterprise (MSE) Lending". *Consultative Group to Assist the Poor (CGAP)*, Washington, DC, USA, 2022. [Online]. Available: <https://www.cgap.org/sites/default/files/publications/Leveraging%20Transactional%20Data.pdf> [Accessed: Oct. 1, 2025].
- [13] A. I. Hashi, "Machine learning-driven fintech solutions for credit scoring and financial inclusion in the gig economy," *J. Financial Innov.*, 2025. [Online]. Available: <https://doi.org/10.38124/ijisrt/25aug1023>
- [14] T. Aslam and A. Aslam, "Social-Credit+: AI-driven social media credit scoring platform," *arXiv preprint arXiv:2506.12099*, 2025. [Online]. Available: <https://doi.org/10.48550/arXiv.2506.12099>.
- [15] V. Garousi, M. Felderer, and M. V. Mäntylä, "Guidelines for including grey literature and conducting multivocal literature reviews in software engineering," *Inf. Softw. Technol.*, vol. 106, pp. 101–121, Feb. 2019. [Online]. Available: <https://doi.org/10.1016/j.infsof.2018.09.006>.
- [16] M. ElSheikh, D. Draheim, and K. van Overveld, "Blockchain-enabled decentralized identity management: The case of self-sovereign identity in public transportation," *J. King Saud Univ. Comput. Inf. Sci.*, vol. 33, no. 8, pp. 944–952, Oct. 2021. [Online]. Available: <https://doi.org/10.1016/j.bcra.2021.100014>.
- [17] T. McConaghy, "The Web3 data economy: Assets, data tokens, and the data value chain," *Ocean Protocol Blog*, 2020, updated 2024. [Online]. Available: <https://oceanprotocol.com/blog/the-web3-data-economy-assets-data-tokens-and-the-data-value-chain> [Accessed: Sep. 25, 2025].
- [18] Insight Genesis, "Decentralized insight generation documentation," 2024. [Online]. Available: <https://docs.insightgenesis.ai/> [Accessed: Sep. 25, 2025].
- [19] Civic Technologies, "Civic Pass: On-chain permissioning for dApps," 2024. [Online]. Available: <https://www.civic.com/civic-pass/> [Accessed: Oct. 5, 2025].

- [20] World Wide Web Consortium (W3C), “Decentralized Identifiers (DIDs) v1.0,” *W3C Recommendation*, 2022. [Online]. Available: <https://www.w3.org/TR/did-core/>
- [21] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001. [Online]. Available: <https://link.springer.com/article/10.1023/A:1010933404324>
- [22] J. Jagtiani and C. Lemieux, “The use of alternative data and machine learning in fintech lending,” *Journal of Banking & Finance*, 2019. [Online]. Available: <https://www.sciencedirect.com/science/article/pii/S037842661830172X>
- [23] VantageScore Solutions, LLC, “Standardizing the industry: The 300–850 range.” [Online]. Available: <https://vantagescore.com/lenders/why-vantagescore/standardized-300-850-range/> [Accessed Jan. 23, 2025].
- [24] Fair Isaac Corporation (FICO), “Understanding risk tiers for enterprise underwriting.” [Online]. Available: <https://www.fico.com/en/products/fico-score> [Accessed Dec. 23, 2025].
- [25] C. Mazzocca *et al.*, “A survey on decentralized identifiers and verifiable credentials,” *IEEE Communications Surveys & Tutorials*, vol. 27, no. 1, 2025. [Online]. Available: <https://doi.org/10.1109/COMST.2025.3543197>
- [26] MIT Registrar’s Office, “Digital diplomas,” 2024. [Online]. Available: <https://registrar.mit.edu/transcripts-records/diplomas/digital-diplomas> [Accessed Jan. 25, 2026].
- [27] Everledger, “Blockchain luxury goods: Helping to improve trust and sustainability,” 2024. [Online]. Available: <https://everledger.io/blockchain-luxury-goods-helping-to-improve-trust-and-sustainability/> [Accessed Jan. 23, 2026].
- [28] M. Sporny *et al.*, “Verifiable credentials data model v2.0,” *W3C Recommendation*, Jan. 2025. [Online]. Available: <https://www.w3.org/TR/vc-data-model-2.0/>
- [29] A. Berke, R. Mahari, S. Pentland, K. Larson, and D. Calacci, “Insights from an experiment crowdsourcing data from thousands of US Amazon users: The importance of transparency, money, and data use,” *Proc. ACM Hum.-Comput. Interact.*, vol. 8, no. CSCW2, Art. no. 466, Nov. 2024. [Online]. Available: <https://doi.org/10.1145/3687005>
- [30] Y. Liang *et al.*, “Decentralized crowdsourcing for human intelligence tasks with efficient on-chain cost,” *Proc. VLDB Endow.*, vol. 15, no. 9, pp. 1875–1888, May 2022. [Online]. Available: [10.1109/ACCESS.2024.3401076](https://doi.org/10.1109/ACCESS.2024.3401076).

- [31] Y. Liang, Y. Li, and B.-S. Shin, "Decentralized crowdsourcing for human intelligence tasks with efficient on-chain cost," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1875–1888, May 2022. [Online]. Available: <https://doi.org/10.14778/3538598.3538609>.
- [32] V. Buterin, "EIP-170: Contract code size limit," *Ethereum Improvement Proposals*, no. 170, Nov. 2016. [Online]. Available: <https://eips.ethereum.org/EIPS/eip-170>
- [33] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, Oct. 2011. [Online]. Available: <https://jmlr.csail.mit.edu/papers/v12/pedregosa11a.html> [Accessed Mar. 15, 2026].
- [34] H. Hofmann, "Statlog (German Credit Data)," *UCI Machine Learning Repository*, 1994. [Online]. Available: <https://archive.ics.uci.edu/dataset/144/statlog+german+credit+data> [Accessed Mar. 19, 2026].
- [35] T. Soni, "Travel Insurance Prediction Data," *Kaggle*, 2021. [Online]. Available: <https://www.kaggle.com/datasets/tejashvi14/travel-insurance-prediction-data>. [Accessed: Mar. 19, 2026].
- [36] H. Chen *et al.*, "Machine Learning Model Evaluation in Smart Contracts," *IEEE Access*, vol. 9, pp. 115241–115252, 2021. [Online]. Available: <https://doi.org/10.1109/ACCESS.2021.3105006>